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Coherently Coupled Mixtures of Bose One-Dimensional Gases

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Abstract

We theoretically investigate the ground-state properties of one-dimensional coherently coupled Bose mixtures with repulsive interactions at zero temperature in the continuum. Previous mean-field treatments suggest that these systems exhibit a polarizing $\mathbb{Z}_2$ symmetry-breaking quantum phase transition, qualitatively similar to that of the Ising model. To study this transition beyond approximate methods, we develop a Diffusion Monte Carlo (DMC) framework incorporating importance sampling, forward walking and an additional spin-flip sampling procedure.

We validate our DMC results quantitatively by comparing the energy density and mean-square order parameter with the exact solution of the two-body Rabi-coupled problem, as well as with mean-field and beyond-mean-field theories in the weakly interacting regime. Excellent agreement is obtained in all cases.

To study the ferromagnetic quantum phase transition, we employ finite-size scaling techniques inspired by classical Ising systems, focusing on peaks in order-parameter fluctuations and Binder-cumulant crossings. We find progressive sharpening of the finite-size crossover as the number of particles is increased, providing evidence for a phase transition occurring in the thermodynamic limit. Small-momentum divergence in the spin static structure factors further demonstrates that the transition originates from an instability in the spin sector.

As the interaction strength increases, we observe shifts in the critical inter-species interaction strength with respect to the mean-field prediction. These shifts are already present in the beyond-mean-field regime and grow progressively with increasing interaction strength, with the unpolarized phase becoming increasingly favored in the strongly interacting regime. The mean-field critical prediction is recovered in the weakly interacting limit. We further find that the mean-field prediction for the critical inter-species interaction remains valid in the ultra-weak Rabi-coupling regime for arbitrary interaction strengths.

Resumen

En este trabajo se estudian teóricamente las propiedades del estado fundamental en gases de bosones unidimensionales de dos componentes con acoplamiento coherente, con interacciones repulsivas, a temperatura cero y en el continuo. Según teoría de campo medio, estos sistemas presentan una transición de fase cuántica de ruptura de simetría $\mathbb{Z}_2$ asociada a la polarización, cualitativamente similar a la transición del modelo de Ising. Para estudiar esta transición más allá de los métodos aproximados, desarrollamos un algoritmo basado en *Diffusion Monte Carlo* (DMC) que incorpora muestreo por importancia, estimadores puros y un procedimiento adicional en el que muestreamos estocásticamente cambios de componente.

Validamos cuantitativamente nuestros resultados de DMC comparando la densidad de energía y el valor cuadrático medio del parámetro de orden con la solución exacta del problema de dos cuerpos con acoplamiento de Rabi, así como con las teorías de campo medio y teoría perturbativa de Bogoliubov en el régimen de interacciones débiles. Se obtiene un excelente acuerdo en todos los casos.

Para validar cualitativamente la transición de fase cuántica ferromagnética, empleamos técnicas de escalado de tamaño finito inspiradas en el modelo de Ising clásico, centrándonos en picos de las fluctuaciones del parámetro de orden y en cruces del cumulante de Binder. Nuestros resultados muestran un progresivo estrechamiento del crossover asociado al tamaño finito a medida que aumenta el número de partículas, proporcionando evidencia de una transición de fase en el límite termodinámico. Los factores de estructura estáticos muestran además que la transición se origina por una inestabilidad en el sector de espín.

A medida que aumenta la intensidad de las interacciones, observamos desplazamientos del valor crítico de la interacción entre especies con respecto a la predicción de campo medio. Estos desplazamientos ya están presentes en el régimen perturbativo de interacciones débiles y aumentan progresivamente con la intensidad de las interacciones, haciendo que la fase no polarizada sea cada vez más favorecida en el régimen fuertemente interactuante. La predicción de campo medio para el valor crítico se recupera en el límite de interacciones débiles. Finalmente, encontramos que la predicción de campo medio para la interacción crítica entre especies sigue siendo válida en el régimen de acoplamiento de Rabi ultradébil para cualquier intensidad de interacción.

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Chapter 1

Problem Statement

Ultracold atomic gases [1] provide a highly controllable platform for exploring quantum many-body physics, where interactions, dimensionality and internal degrees of freedom can be tuned with remarkable precision [2]. In particular, one-dimensional bosonic systems [3] display exotic quantum phenomena as strong fluctuations prevent true long-range order [4], while strong interactions drive the system to a direct mapping onto fermionic behavior [5]. Extending these 1D systems to include internal degrees of freedom unlocks a richer landscape of quantum phenomena [6]. Experimentally, such quantum-degenerate mixtures can be realized with atomic gases occupying different hyperfine states [7, 8] or with distinct atomic species [9]. Furthermore, coherently coupled configurations can be realized through radio-frequency transitions between internal states [10]. When conversion between internal states is allowed, the interplay between coherence and interactions can give rise to magnetic ground-state phases and a second-order quantum phase transition [11], characterized by the breaking of a $\mathbb{Z}_2$ symmetry between paramagnetic and ferromagnetically ordered configurations, reminiscent of the classical Ising model transition [12, 13].

Quantum Monte Carlo methods [14, 15] are well suited to describe these systems. In particular, the Diffusion Monte Carlo (DMC) method [15] allows one to quantitatively describe these quantum systems exactly, but incorporating internal-state conversion remains challenging, and no standard techniques currently exist. Therefore, to the best of our knowledge, coherently coupled mixtures of Bose gases have only been studied theoretically via mean-field [16, 17] and Bogoliubov [18, 19, 20] approaches. This thesis aims to implement and validate internal-state transition processes within DMC simulations for mixtures of one-dimensional Bose gases and to study their ground-state properties and critical aspects using a classical approach given the resemblance with an Ising transition.

The remainder of this thesis is organized as follows. In chapter 2, we discuss the fundamental results concerning one- and two-component one-dimensional Bose gases, followed by the state-of-the-art theory used to describe coherently coupled mixtures. In chapter 3, we deeply explore the Diffusion Monte Carlo method and develop a proper framework to simulate the system of interest. Finally, in chapter 4, we validate the Monte Carlo results by comparing them against the previously introduced theoretical approaches. The conclusions of this thesis are presented in Chapter 5.

Chapter 2

Theoretical Background

This chapter sets the theoretical foundations for one- and two-component one-dimensional Bose gases and reviews the current state of research on coherently-coupled Bose mixtures. We conclude with the exact solution of the two-body problem.

2.1 One-Dimensional Bose Gases

The theoretical framework for one-dimensional Bose gases at zero temperature ($T = 0$) is discussed in this section. We focus exclusively on continuum systems with repulsive, contact (point-like) interactions, setting aside lattice models, external trapping potentials, and attractive regimes. While not relevant for many qualitative results discussed in the following sections, we consider a ring geometry, which leads to periodic boundary conditions in the quantitative description that follows.

2.1.1 Single-Component One-Dimensional Bose Gases

One-dimensional Bose gases with one component have been actively investigated both theoretically and experimentally [21, 22, 23]. For ultracold alkali atoms, the interaction potential can be well described by a delta interaction and, in one dimension, corresponds to the Lieb-Liniger model, represented by the following Hamiltonian:

$$\hat{H} = -\frac{\hbar^2}{2m} \sum_{i=1}^N \frac{\partial^2}{\partial x_i^2} + g \sum_{i<j}^N \delta(x_i - x_j), \quad x_i \in [0, L), \quad (2.1)$$

where we have assumed all involved bosons have the same mass m and they all interact with the same strength g . For repulsive interactions, the coupling constant is positive, $g > 0$. We consider a finite system containing a total number of particles N on a ring of length L , or, equivalently, in a box of length L in the presence of periodic boundary conditions.

It is common to parametrize the interaction strength using the dimensionless (Lieb-Liniger) parameter $\gamma = \frac{mg}{\hbar^2 n}$, where $n = N/L$ denotes the linear density. The behavior of this system is dictated by γ , separating into two main physical regimes. In the weakly interacting regime ($\gamma \ll 1$), the system is coherent which gives rise to a quasi-Bose-Einstein condensate. In this weakly interacting limit, the system is well described by Gross-Pitaevskii

and Bogoliubov perturbative theories (see, for example [24]), with $\gamma = 0$ representing non-interacting bosons. In contrast, the strongly interacting regime ($\gamma \gg 1$) occurs at low density where interactions dominate, leading to local anti-bunching. As $\gamma \rightarrow \infty$, the gas reaches the Tonks-Girardeau [5] limit of hard-core bosons, where the model can be solved exactly by mapping the wavefunctions onto a system of non-interacting spinless fermions. Moreover, the model can be solved for all values of γ using the Bethe ansatz [21, 25].

2.1.2 Mixtures of One-Dimensional Bose Gases

Even richer physics emerges when a binary mixture of two one-dimensional Bose gases is considered [6]. In this framework, particles possess two possible internal states, which can be labeled as $\{\uparrow, \downarrow\}$. We assume that two hyperfine states of the same atomic species are considered, such that both components share an identical particle mass m . The single-component Hamiltonian (2.1) can be straightforwardly extended to this two-component paradigm as

$$\begin{aligned} \hat{H} = & -\frac{\hbar^2}{2m} \left(\sum_{i=1}^{N_\uparrow} \frac{\partial^2}{\partial x_i^2} + \sum_{j=1}^{N_\downarrow} \frac{\partial^2}{\partial y_j^2} \right) + g_{\uparrow\uparrow} \sum_{1 \leq i < j \leq N_\uparrow} \delta(x_i - x_j) \\ & + g_{\downarrow\downarrow} \sum_{1 \leq i < j \leq N_\downarrow} \delta(y_i - y_j) + g_{\uparrow\downarrow} \sum_{i=1}^{N_\uparrow} \sum_{j=1}^{N_\downarrow} \delta(x_i - y_j), \quad x_i, y_j \in [0, L), \end{aligned} \quad (2.2)$$

where $g_{\uparrow\uparrow}$ and $g_{\downarrow\downarrow}$ denote the intra-species coupling constants for components $\uparrow$ and $\downarrow$, such that $g_{\uparrow\downarrow}$ represents the inter-species interaction strength, and $N_\uparrow, N_\downarrow$ are the respective particle numbers. In general, model (2.2) is non-integrable. For the symmetric mixture ($g_{\uparrow\uparrow} = g_{\downarrow\downarrow} = g$), interactions can be parametrized by the dimensionless quantities $\gamma = \frac{mg}{\hbar^2 n}$ and $\gamma_{\uparrow\downarrow} = \frac{mg_{\uparrow\downarrow}}{\hbar^2 n}$, defined via the total linear density $n = (N_\uparrow + N_\downarrow)/L$. The analogous weakly and strongly interacting regimes correspond to $\gamma, \gamma_{\uparrow\downarrow} \ll 1$ and $\gamma, \gamma_{\uparrow\downarrow} \gg 1$, respectively.

In two-component one-dimensional Bose gases, low-energy dynamics can be understood by separating the degrees of freedom into density and spin sectors [24]. The density sector describes in-phase oscillations of the total fluid, representing acoustic compressional waves. The spin sector captures out-of-phase oscillations, where the two components flow relative to one another while keeping the total density constant.

A remarkably interesting feature of one-dimensional Bose mixtures is related to the demixing phase transition [24, 26]. Driven by the competition between intra- and inter-species interactions, the system shifts from a miscible phase, where component mixing is energetically favored, to a phase-separated state where particles segregate into single-component spatial domains. In a symmetric repulsive mixture, this quantum phase transition occurs at $g_{\uparrow\downarrow} = g$. Although originally predicted via mean-field theory, this first-order quantum phase transition remains robust across all finite interaction strengths*[27].

*The sole exception is the extreme Tonks-Girardeau limit ($\gamma \rightarrow \infty$), which maps exactly onto a two-component Fermi gas with repulsion that remains permanently miscible.

2.2 Coherently-Coupled Bose Mixtures

In this section, we introduce the Rabi-coupled model and analyze its mean-field ground state. We also discuss the Bogoliubov excitation spectrum, beyond-mean-field corrections and exact two-body solution following recent work by D.S. Petrov [28].

2.2.1 Rabi-Coupled Model

The Hamiltonian of our Rabi-coupled one-dimensional mixture corresponds to the symmetric version of Eq. (2.2) supplemented by a tunneling term that drives tunneling between internal states, namely:

$$\begin{aligned} \hat{H} = & -\frac{\hbar^2}{2m} \left(\sum_{i=1}^{N_\uparrow} \frac{\partial^2}{\partial x_i^2} + \sum_{j=1}^{N_\downarrow} \frac{\partial^2}{\partial y_j^2} \right) + g \sum_{1 \leq i < j \leq N_\uparrow} \delta(x_i - x_j) \\ & + g \sum_{1 \leq i < j \leq N_\downarrow} \delta(y_i - y_j) + g_{\uparrow\downarrow} \sum_{i=1}^{N_\uparrow} \sum_{j=1}^{N_\downarrow} \delta(x_i - y_j) - t \sum_{k=1}^N \hat{X}_k, \quad x_i, y_j \in [0, L), \end{aligned} \quad (2.3)$$

where t quantifies the strength of the Rabi coupling (single-particle tunneling rate) and N is the total particle number. For later use, we define the dimensionless tunneling parameter $\tilde{t} = \frac{mt}{\hbar^2 n^2}$. The Rabi term acts via single-particle spin-flip operators $\hat{X}_k$. Acting on the many-body wavefunction $\Phi(x_1, \sigma_1; \dots; x_k, \sigma_k; \dots; x_N, \sigma_N)$, the operator $\hat{X}_k$ flips the internal spin state argument $\sigma_k \in \{\uparrow, \downarrow\}$ while preserving all spatial coordinates:

$$\hat{X}_k \Phi(x_1, \sigma_1; \dots; x_k, \sigma_k; \dots; x_N, \sigma_N) = \Phi(x_1, \sigma_1; \dots; x_k, \bar{\sigma}_k; \dots; x_N, \sigma_N), \quad (2.4)$$

where $\bar{\sigma}_k$ denotes the inverted spin state ($\bar{\uparrow} = \downarrow$ and $\bar{\downarrow} = \uparrow$). While Hamiltonian (2.2) conserves $N_\uparrow$ and $N_\downarrow$ independently, the coherent coupling in (2.3) explicitly breaks this symmetry, conserving only the total particle number N . Model (2.3) has been extensively studied using approximate methods in the weakly-interacting regime. To provide a convenient foundation for both mean-field (MF) and beyond-mean-field (BMF) treatments, it is instructive to reframe model (2.3) in second quantization within the grand-canonical ensemble. In this formalism, the system is described by continuous bosonic field operators $\hat{\Psi}_\sigma(x)$ and $\hat{\Psi}_\sigma^\dagger(x)$, which annihilate and create a particle of component $\sigma \in \{\uparrow, \downarrow\}$ at position x , respectively, satisfying equal-time commutation relations $[\hat{\Psi}_\sigma(x), \hat{\Psi}_\sigma^\dagger(x')] = \delta_{\sigma\sigma'} \delta(x - x')$ [24]. Thermodynamic equilibrium is governed by the full grand potential operator $\hat{\Omega} = \hat{H} - \mu_\uparrow \hat{N}_\uparrow - \mu_\downarrow \hat{N}_\downarrow$ (μ_σ denoting the chemical potential for component σ), which takes the explicit form

$$\begin{aligned} \hat{\Omega} = & \int_0^L dx \left[\sum_{\sigma \in \{\uparrow, \downarrow\}} \hat{\Psi}_\sigma^\dagger(x) \left(-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} - \mu_\sigma \right) \hat{\Psi}_\sigma(x) + \frac{g}{2} \sum_{\sigma \in \{\uparrow, \downarrow\}} \hat{\Psi}_\sigma^\dagger(x) \hat{\Psi}_\sigma^\dagger(x) \hat{\Psi}_\sigma(x) \hat{\Psi}_\sigma(x) \right. \\ & \left. + g_{\uparrow\downarrow} \hat{\Psi}_\uparrow^\dagger(x) \hat{\Psi}_\downarrow^\dagger(x) \hat{\Psi}_\downarrow(x) \hat{\Psi}_\uparrow(x) - t \left(\hat{\Psi}_\uparrow^\dagger(x) \hat{\Psi}_\downarrow(x) + \hat{\Psi}_\downarrow^\dagger(x) \hat{\Psi}_\uparrow(x) \right) \right], \end{aligned} \quad (2.5)$$

2.2.2 Mean-Field Ground State

Within the grand-canonical field formalism, the mean-field description corresponds to evaluating the grand potential operator (2.5) at zeroth order in quantum fluctuations, replacing the field operators with classical, spatially uniform order parameters $\hat{\Psi}_\sigma(x) \rightarrow \psi_\sigma = \sqrt{n_\sigma} e^{i\phi_\sigma}$ ($\sigma \in \{\uparrow, \downarrow\}$). For a homogeneous system in the phase-locked ground state ($\phi_\uparrow - \phi_\downarrow = 0$), spatial derivatives vanish, and the zeroth-order grand potential density reduces to $\Omega^{(0)}/L = E_{\text{MF}}/L - \mu_\uparrow n_\uparrow - \mu_\downarrow n_\downarrow$. Here, E_{MF}/L is the classical mean-field energy per unit length:

$$\frac{E_{\text{MF}}}{L} = \frac{1}{2}g(n_\uparrow^2 + n_\downarrow^2) + g_{\uparrow\downarrow}n_\uparrow n_\downarrow - 2t\sqrt{n_\uparrow n_\downarrow}, \quad (2.6)$$

where $n_\uparrow = |\psi_\uparrow|^2$ and $n_\downarrow = |\psi_\downarrow|^2$ are the component particle densities. To analyze the ground state, it is convenient to express (2.6) in terms of the total density $n = n_\uparrow + n_\downarrow$ and the polarization $P = (n_\uparrow - n_\downarrow)/n$:

$$\frac{E_{\text{MF}}(P)}{L} = \frac{n^2}{4} \left[(g + g_{\uparrow\downarrow}) + (g - g_{\uparrow\downarrow})P^2 - \frac{4t}{n}\sqrt{1 - P^2} \right]. \quad (2.7)$$

Minimizing $E_{\text{MF}}(P)$ with respect to P reveals two distinct ground-state regimes (Fig. 2.1, a) depending on the interaction and tunneling strengths: an unpolarized ground state ($P = 0$) for $g_{\uparrow\downarrow} < g + 2t/n$, where the two components remain fully mixed with equal densities $n_\uparrow = n_\downarrow = n/2$, and a polarized ground state ($P \neq 0$) for $g_{\uparrow\downarrow} > g + 2t/n$, where the system spontaneously breaks $\mathbb{Z}_2$ symmetry to yield a non-zero polarization, namely:

$$P = \begin{cases} 0, & \text{for } g_{\uparrow\downarrow} < g + \frac{2t}{n}, \\ \pm \sqrt{1 - \left[\frac{2t}{n(g - g_{\uparrow\downarrow})} \right]^2}, & \text{for } g_{\uparrow\downarrow} > g + \frac{2t}{n}. \end{cases} \quad (2.8)$$

The threshold $g_{\uparrow\downarrow}^{\text{MF}} = g + 2t/n$ defines the critical point of a continuous second-order quantum phase transition in which the polarization P serves as the order parameter.

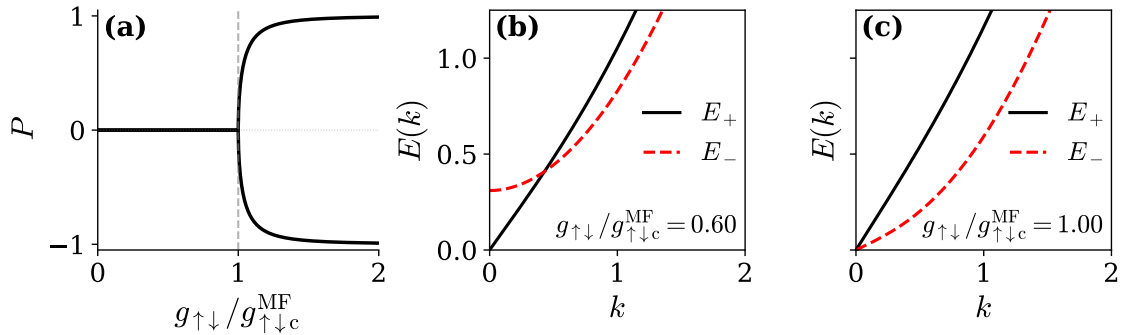


Figure 2.1: (a) Ground state polarization as a function of $g_{\uparrow\downarrow}/g_{\uparrow\downarrow}^{\text{MF}}$. Dashed line shows the critical point. (b) Dispersion relation above unpolarized ground state for $g_{\uparrow\downarrow} = 0.6g_{\uparrow\downarrow}^{\text{MF}}$. (c) Dispersion relation at the critical point. $t/g = 0.1$. Here $E(k)$ is measured in units of gn , and k in units of $\hbar/\sqrt{mgn}$, which is proportional to the healing length.

2.2.3 Bogoliubov Excitations

To analyze the excitation spectrum and the nature of the quantum phase transition, we expand the Bose field operators around the mean-field ground state:

$$\hat{\Psi}_\sigma(x, t) \approx e^{-i\mu_\sigma t/\hbar} \left[\sqrt{n_\sigma} + \hat{\phi}_\sigma(x, t) \right], \quad \sigma \in \{\uparrow, \downarrow\}, \quad (2.9)$$

where n_σ are the homogeneous condensate densities, μ_σ are the mean-field chemical potentials, and $\hat{\phi}_\sigma(x, t)$ represent small quantum fluctuations. Expanding the system Hamiltonian up to quadratic order in $\hat{\phi}_\sigma$ yields the Bogoliubov Hamiltonian:

$$\hat{H}_2 = \frac{1}{2} \sum_k \begin{pmatrix} \hat{\phi}_{\uparrow,k}^\dagger & \hat{\phi}_{\downarrow,k}^\dagger & \hat{\phi}_{\uparrow,-k} & \hat{\phi}_{\downarrow,-k} \end{pmatrix} \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B} & \mathbf{A} \end{pmatrix} \begin{pmatrix} \hat{\phi}_{\uparrow,k} \\ \hat{\phi}_{\downarrow,k} \\ \hat{\phi}_{\uparrow,-k}^\dagger \\ \hat{\phi}_{\downarrow,-k}^\dagger \end{pmatrix}, \quad (2.10)$$

where $\hat{\phi}_{\sigma,k}$ is the annihilation operator for a single-particle mode of momentum k . For a symmetric mixture ($g_{\uparrow\uparrow} = g_{\downarrow\downarrow} = g$), the submatrices $\mathbf{A}$ and $\mathbf{B}$ take the explicit form:

$$\mathbf{A} = \begin{pmatrix} \varepsilon_k + gn_\uparrow + t\sqrt{\frac{n_\downarrow}{n_\uparrow}} & g_{\uparrow\downarrow}\sqrt{n_\uparrow n_\downarrow} - t \\ g_{\uparrow\downarrow}\sqrt{n_\uparrow n_\downarrow} - t & \varepsilon_k + gn_\downarrow + t\sqrt{\frac{n_\uparrow}{n_\downarrow}} \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} gn_\uparrow & g_{\uparrow\downarrow}\sqrt{n_\uparrow n_\downarrow} \\ g_{\uparrow\downarrow}\sqrt{n_\uparrow n_\downarrow} & gn_\downarrow \end{pmatrix}, \quad (2.11)$$

with $\varepsilon_k = \frac{\hbar^2 k^2}{2m}$ being the free-particle kinetic energy. The quadratic Hamiltonian (2.10) is diagonalized by applying a canonical linear Bogoliubov transformation that relates the field fluctuation operators $\hat{\phi}_{\sigma,k}$ to non-interacting quasiparticle operators $\hat{b}_{\lambda,k}$ ($\lambda \in \{+, -\}$):

$$\begin{pmatrix} \hat{\phi}_{\uparrow,k} \\ \hat{\phi}_{\downarrow,k} \end{pmatrix} = \sum_{\lambda \in \{+, -\}} \left[\mathbf{u}^{(\lambda)}(k) \hat{b}_{\lambda,k} + \mathbf{v}^{(\lambda)*}(-k) \hat{b}_{\lambda,-k}^\dagger \right], \quad (2.12)$$

where $\mathbf{u}^{(\lambda)} = (u_\uparrow^{(\lambda)}, u_\downarrow^{(\lambda)})^T$ and $\mathbf{v}^{(\lambda)} = (v_\uparrow^{(\lambda)}, v_\downarrow^{(\lambda)})^T$ are the component transformation amplitudes. These amplitudes and the quasiparticle excitation spectrum $E_{\lambda,k}$ are obtained by solving the non-Hermitian eigenvalue problem:

$$\begin{pmatrix} \mathbf{A} & \mathbf{B} \\ -\mathbf{B}^* & -\mathbf{A}^* \end{pmatrix} \begin{pmatrix} u_\uparrow^{(\lambda)} \\ u_\downarrow^{(\lambda)} \\ v_\uparrow^{(\lambda)} \\ v_\downarrow^{(\lambda)} \end{pmatrix} = E_{\lambda,k} \begin{pmatrix} u_\uparrow^{(\lambda)} \\ u_\downarrow^{(\lambda)} \\ v_\uparrow^{(\lambda)} \\ v_\downarrow^{(\lambda)} \end{pmatrix}, \quad (2.13)$$

subject to the bosonic normalization condition $|u_\uparrow^{(\lambda)}|^2 + |u_\downarrow^{(\lambda)}|^2 - |v_\uparrow^{(\lambda)}|^2 - |v_\downarrow^{(\lambda)}|^2 = 1$. In the unpolarized symmetric phase ($n_\uparrow = n_\downarrow = n/2$), solving Eq. (2.13) shows that the two modes completely decouple into total-density (+) and spin (−) fluctuations, yielding the

two excitation branches:

$$E_{+,k} = \sqrt{\varepsilon_k (\varepsilon_k + (g + g_{\uparrow\downarrow})n)}, \quad (2.14)$$

$$E_{-,k} = \sqrt{(\varepsilon_k + 2t) (\varepsilon_k + (g - g_{\uparrow\downarrow})n + 2t)}. \quad (2.15)$$

The upper branch, $E_{+,k}$, corresponds to gapless total-density (sound) fluctuations, exhibiting a linear acoustic dispersion $E_{+,k} \approx c_d k$ at long wavelengths ($k \rightarrow 0$) with sound velocity $c_d = \sqrt{(g + g_{\uparrow\downarrow})n/2m}$. In contrast, the lower branch, $E_{-,k}$, represents relative-density (spin) fluctuations. Due to the coherent Rabi coupling t , this spin mode acquires a finite energy gap at $k = 0$ (Figure 2.1, b), defining the Josephson frequency $\hbar\omega_J = \sqrt{2t[(g - g_{\uparrow\downarrow})n + 2t]}$. As the interspecies interaction $g_{\uparrow\downarrow}$ increases toward the critical mean-field threshold $g_{\uparrow\downarrow}^{\text{MF}} = g + 2t/n$, the spin gap softens to zero (Figure 2.1, c) which confirms that the quantum phase transition is driven purely by an instability in the spin sector.

2.2.4 Beyond-Mean-Field Corrections

In this section, we utilize Bogoliubov theory to derive second-order quantum corrections to the ground-state energy density and the mean-square polarization. See Appendix A for the general Bogoliubov excitation spectrum.

Beyond-Mean-Field Energy Corrections

To incorporate quantum fluctuations beyond the mean-field approximation, the ground-state energy density is expanded up to second order in the fluctuation fields. The ground-state energy per unit length at the Bogoliubov level is given by

$$\frac{E_{\text{total}}}{L} = \frac{E_{\text{MF}}}{L} + \frac{E_{\text{LHY}}}{L} + \dots, \quad (2.16)$$

where E_{MF}/L is the mean-field energy density, and E_{LHY}/L represents the Lee-Huang-Yang (LHY) quantum correction [29, 30]. In one dimension, the LHY energy density is obtained by summing the zero-point quantum energies of both Bogoliubov excitation branches over all wavevectors k :

$$\frac{E_{\text{LHY}}}{L} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{dk}{2\pi} \left(E_{+,k} + E_{-,k} - 2\varepsilon_k - \frac{2t(n_{\uparrow} + n_{\downarrow})}{\sqrt{n_{\uparrow}n_{\downarrow}}} - g_{\uparrow\uparrow}n_{\uparrow} - g_{\downarrow\downarrow}n_{\downarrow} \right), \quad (2.17)$$

where $E_{+,k}$ and $E_{-,k}$ are the density- and spin-like excitation branches. To account for finite-system effects with periodic boundary conditions, the continuous momentum integral in Eq. (2.17) is replaced by a discrete sum over wavevectors $k_m = 2\pi m/L$:

$$\frac{E_{\text{BMF}}^{(L)}}{L} = \frac{1}{L} \left[\frac{1}{2} \mathcal{I}(0) + \sum_{m=1}^{\infty} \mathcal{I}(k_m) \right], \quad (2.18)$$

where $\mathcal{I}(k)$ is the integrand from Eq. (2.17).

Beyond-Mean-Field Order Parameter Corrections

At the mean-field level, the order parameter is given by the classical ground-state polarization (2.8). Beyond mean-field, quantum zero-point corrections influence the order parameter in two distinct physical ways: through a macroscopic thermodynamic shift, δP_{LHY} , driven by the effective pressure of the Lee-Huang-Yang energy, and through intrinsic quantum fluctuations, $\langle \delta \hat{P}^2 \rangle$, from $k = 0$ zero-point spin-flip excitations.

The LHY energy correction E_{LHY}/L perturbs the chemical potentials, inducing density shifts $n_\sigma \rightarrow n_\sigma + \delta n_\sigma$. Using grand-canonical thermodynamics, these corrections relate to the LHY force and a uniform chemical potential shift $\delta\mu$ by:

$$\delta n_\sigma = - \sum_{\sigma'} (\mathcal{M}^{-1})_{\sigma\sigma'} \frac{\partial(E_{\text{LHY}}/L)}{\partial n_{\sigma'}} + \delta\mu \sum_{\sigma'} (\mathcal{M}^{-1})_{\sigma\sigma'}, \quad (2.19)$$

where $\mathcal{M}_{\sigma\sigma'} = \frac{\partial^2(E_{\text{MF}}/L)}{\partial n_\sigma \partial n_{\sigma'}} = \frac{\partial \mu_\sigma}{\partial n_{\sigma'}}$ is the symmetric mean-field compressibility matrix, defined explicitly as:

$$\mathcal{M} = \begin{pmatrix} g + \frac{t}{2} \frac{\sqrt{n_\downarrow}}{n_\uparrow^{3/2}} & g_{\uparrow\downarrow} - \frac{t}{2} \frac{1}{\sqrt{n_\uparrow n_\downarrow}} \\ g_{\uparrow\downarrow} - \frac{t}{2} \frac{1}{\sqrt{n_\uparrow n_\downarrow}} & g + \frac{t}{2} \frac{\sqrt{n_\uparrow}}{n_\downarrow^{3/2}} \end{pmatrix}. \quad (2.20)$$

Enforcing fixed total particle density $n = n_\uparrow + n_\downarrow$ ($\delta n_\uparrow = -\delta n_\downarrow$) uniquely fixes the chemical potential shift:

$$\delta\mu = \frac{\sum_{\sigma,\sigma'} (\mathcal{M}^{-1})_{\sigma\sigma'} \frac{\partial(E_{\text{LHY}}/L)}{\partial n_{\sigma'}}}{\sum_{\sigma,\sigma'} (\mathcal{M}^{-1})_{\sigma\sigma'}}. \quad (2.21)$$

Substituting $\delta\mu$ back into $\delta n_\uparrow$ yields the BMF correction to the mean polarization order parameter, $P = P_{\text{MF}} + \delta P_{\text{LHY}}$, where:

$$\delta P_{\text{LHY}} = \frac{\delta n_\uparrow - \delta n_\downarrow}{n} = \frac{2\delta n_\uparrow}{n}. \quad (2.22)$$

In finite systems, zero-mode ($k = 0$) excitations conserve N by exchanging particles exclusively via out-of-phase spin flips ($\delta \hat{N}_\uparrow = -\delta \hat{N}_\downarrow$). Evaluating the variance of the polarization $\hat{P} = (\hat{N}_\uparrow - \hat{N}_\downarrow)/N$ using the $k = 0$ amplitudes defined in Eq. (2.12) yields the Gaussian quantum noise:

$$\langle \delta \hat{P}^2 \rangle = \frac{4N_\uparrow}{N^2} \left| u_\uparrow^{(+)} + v_\uparrow^{(+)} \right|^2 \quad (2.23)$$

where $u_\sigma^{(+)}$ and $v_\sigma^{(+)}$ correspond to the upper spin-excitation branch evaluated at $k = 0$. Combining both the thermodynamic shift and zero-point quantum noise yields the total observable mean-square order parameter:

$$\langle \hat{P}^2 \rangle = (P_{\text{MF}} + \delta P_{\text{LHY}})^2 + \langle \delta \hat{P}^2 \rangle. \quad (2.24)$$

2.3 Exact Solution of the Two-Body Problem

For validation purposes, the exact two-body problem in a one-dimensional ring geometry can be solved (see Supplemental Material of [31] for the solution in 3D). In a resonant, driven two-component 1D Bose gas subject to driving amplitude t , single-particle internal states mix via $-t\hat{X}$, yielding energy eigenvalues $\pm t$. The internal single-particle eigenstates reduce to symmetric and antisymmetric superpositions $|+\rangle = |\uparrow\rangle - |\downarrow\rangle$ and $|-\rangle = |\uparrow\rangle + |\downarrow\rangle$.

For two identical bosons, the system decomposes into three symmetric scattering channels separated in energy by $2t$:

$$|--\rangle = |\uparrow\uparrow\rangle + |\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle + |\downarrow\downarrow\rangle, \quad (2.25)$$

$$|+-\rangle = |\uparrow\uparrow\rangle - |\downarrow\downarrow\rangle, \quad (2.26)$$

$$|++\rangle = |\uparrow\uparrow\rangle - (|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle) + |\downarrow\downarrow\rangle. \quad (2.27)$$

The channel kets are unnormalized for algebraic simplicity, with all normalization prefactors absorbed into the spatial expansion coefficients. In the relative coordinate $x = x_1 - x_2$, the even spatial wave function is expressed across these channels as

$$\Psi(x) = \cosh(\kappa x)|--\rangle + C_1 \cosh(\kappa_1 x)|+-\rangle + C_2 \cosh(\kappa_2 x)|++\rangle, \quad (2.28)$$

where $\kappa_1 = \sqrt{4mt/\hbar^2 + \kappa^2}$ and $\kappa_2 = \sqrt{8mt/\hbar^2 + \kappa^2}$. Projecting Eq. (2.28) back onto the original spin basis yields:

$$\Psi_{\uparrow\uparrow}(x) = \cosh(\kappa x) + C_1 \cosh(\kappa_1 x) + C_2 \cosh(\kappa_2 x), \quad (2.29)$$

$$\Psi_{\uparrow\downarrow}(x) = \cosh(\kappa x) - C_2 \cosh(\kappa_2 x), \quad (2.30)$$

$$\Psi_{\downarrow\downarrow}(x) = \cosh(\kappa x) - C_1 \cosh(\kappa_1 x) + C_2 \cosh(\kappa_2 x). \quad (2.31)$$

The coefficients $C_{1,2}$ and momentum parameter κ are uniquely fixed by matching zero-range Bethe-Peierls [32] boundary conditions at $x = \pm L/2$:

$$\frac{mg_{\sigma\sigma'}}{2\hbar^2}\Psi_{\sigma\sigma'}\left(\frac{L}{2}\right) + \partial_x\Psi_{\sigma\sigma'}\left(\frac{L}{2}\right) = 0, \quad \text{for } \sigma, \sigma' \in \{\uparrow, \downarrow\}. \quad (2.32)$$

Measured relative to twice the single-particle ground threshold $-t$, the ground-state energy reads

$$E = -\frac{\hbar^2\kappa^2}{m} - 2t. \quad (2.33)$$

Spatial probabilities $P_{\sigma\sigma'}$ of finding the pair in specific spin configurations are evaluated via

$$P_{\uparrow\uparrow} = A^2 \int_0^{L/2} |\Psi_{\uparrow\uparrow}(x)|^2 dx, \quad P_{\uparrow\downarrow} = 2A^2 \int_0^{L/2} |\Psi_{\uparrow\downarrow}(x)|^2 dx, \quad P_{\downarrow\downarrow} = A^2 \int_0^{L/2} |\Psi_{\downarrow\downarrow}(x)|^2 dx, \quad (2.34)$$

where A is the global normalization constant uniquely determined by enforcing $\sum_{\sigma\sigma'} P_{\sigma\sigma'} = 1$, which absorbs all prefactors arising from the internal channel projection.

Chapter 3

Numerical Method

In this section, we present the numerical framework developed to simulate the system of interest, based on a modified importance-sampled Diffusion Monte Carlo (DMC) algorithm. First, we outline the standard DMC method. We then introduce importance sampling, discuss our choice of trial wavefunction and address the mixed estimator problem. Next, we extend the algorithm to incorporate spin-flip dynamics. We follow by detailing the physical observables used to analyze these effects. We conclude by describing the final DMC algorithm used to simulate the system of interest.

3.1 Standard Diffusion Monte Carlo Method

Diffusion Monte Carlo (DMC) [15] method is a projection quantum Monte Carlo technique designed to extract the exact ground state of a many-body quantum system by propagating the Schrödinger equation in imaginary time. In this section, we review the mathematical foundations of the ordinary DMC algorithm and its formulation as a combined diffusion and branching process. From here on we use notation with $\hbar = m = 1$.

3.1.1 The Imaginary-Time Schrödinger Equation

The Diffusion Monte Carlo mechanism relies on transforming the time-dependent Schrödinger equation to imaginary time via a Wick rotation, $\tau = it_{\text{real}}$, where t_{real} denotes the real time variable. Introducing a reference energy shift E_T , the imaginary-time evolution is governed by

$$-\frac{\partial \phi(\mathbf{x}, \tau)}{\partial \tau} = (H - E_T)\phi(\mathbf{x}, \tau). \quad (3.1)$$

Expanding the initial state in the energy eigenbasis of H , $\phi(\mathbf{x}, 0) = \sum_{k=0}^{\infty} C_k \Phi_k(\mathbf{x})$ with $C_k = \langle \Phi_k | \phi(0) \rangle$, the exact formal solution yields

$$\phi(\mathbf{x}, \tau) = C_0 \Phi_0(\mathbf{x}) e^{-(E_0 - E_T)\tau} + \sum_{k=1}^{\infty} C_k \Phi_k(\mathbf{x}) e^{-(E_k - E_T)\tau}. \quad (3.2)$$

Provided $E_T \approx E_0$ and the initial state has a non-zero overlap with the ground state ($C_0 \neq 0$), all excited-state contributions ($E_k > E_0$) are exponentially suppressed relative

to Φ_0 as $\tau \rightarrow \infty$. Consequently, long imaginary-time propagation filters out higher energy modes and projects the system directly onto its ground state, $\phi(\mathbf{x}, \tau \rightarrow \infty) \propto \Phi_0(\mathbf{x})$.

3.1.2 Green's Function Formalism

To formulate the imaginary-time evolution as a stochastic process, Eq. (3.1) is expressed in integral form via the Green's function formalism:

$$\phi(\mathbf{x}', \tau + \Delta\tau) = \int G(\mathbf{x}', \mathbf{x}; \Delta\tau) \phi(\mathbf{x}, \tau) d\mathbf{x}, \quad (3.3)$$

where the Green's function represents the spatial transition probability density from $\mathbf{x}$ to $\mathbf{x}'$ over an imaginary-time step $\Delta\tau$. In coordinate representation, $G(\mathbf{x}', \mathbf{x}; \Delta\tau)$ corresponds to the matrix element of the imaginary-time evolution operator:

$$G(\mathbf{x}', \mathbf{x}; \Delta\tau) = \langle \mathbf{x}' | e^{-(H-E_T)\Delta\tau} | \mathbf{x} \rangle. \quad (3.4)$$

Repeated application of Eq. (3.3) over n discrete time steps propagates the initial state according to

$$\phi(\mathbf{x}', n\Delta\tau) = \int G(\mathbf{x}', \mathbf{x}_0; n\Delta\tau) \phi(\mathbf{x}_0, 0) d\mathbf{x}_0. \quad (3.5)$$

In the asymptotic limit $n\Delta\tau \rightarrow \infty$, propagating an ensemble of configurations (walkers) according to the transition kernel $G(\mathbf{x}', \mathbf{x}; \Delta\tau)$ causes their spatial distribution to converge to the exact ground-state wavefunction $\Phi_0(\mathbf{x})$.

3.1.3 Diffusion and Branching

The core mechanism of DMC relies on mapping the imaginary-time Schrödinger equation onto a combined classical diffusion and rate (branching) process. Splitting the Hamiltonian into kinetic and potential contributions, $H = T + V$, Eq. (3.1) for an N -body system in 1D reads,

$$-\frac{\partial \phi(\mathbf{x}, \tau)}{\partial \tau} = -D \sum_{i=1}^N \frac{\partial^2 \phi(\mathbf{x}, \tau)}{\partial x_i^2} + [V(\mathbf{x}) - E_T] \phi(\mathbf{x}, \tau), \quad (3.6)$$

where $\mathbf{x} = (x_1, \dots, x_N) \in \mathbb{R}^N$ represents the N -particle spatial configuration, and D , the “diffusion” constant is $\hbar^2/2m = 1/2$ in dimensionless units.

Without the potential term, Eq. (3.6) is an N -dimensional diffusion equation. Without the kinetic term, it reduces to a first-order rate process governed by the local growth/decay rate $[E_T - V(\mathbf{x})]$. To approximate the full propagator over a small time step $\Delta\tau$, we employ the Lie-Trotter primitive approximation, factorizing $e^{-(T+V-E_T)\Delta\tau} \approx e^{-T\Delta\tau} e^{-(V-E_T)\Delta\tau} + \mathcal{O}(\Delta\tau^2)$. In coordinate space, the short-time Green's function factorizes as:

$$G(\mathbf{x}', \mathbf{x}; \Delta\tau) \approx G_{\text{diff}}(\mathbf{x}', \mathbf{x}; \Delta\tau) G_{\text{B}}(\mathbf{x}', \mathbf{x}; \Delta\tau). \quad (3.7)$$

The kinetic contribution yields free diffusion for each particle in 1D, described by an N -dimensional Gaussian kernel:

$$G_{\text{diff}}(\mathbf{x}', \mathbf{x}; \Delta\tau) = (4\pi D\Delta\tau)^{-N/2} \exp \left[-\frac{(\mathbf{x}' - \mathbf{x})^2}{4D\Delta\tau} \right]. \quad (3.8)$$

Stochastically, G_{diff} propagates each particle's position according to independent Gaussian random displacements with variance $\sigma^2 = 2D\Delta\tau$. The potential contribution governs walker branching through the multiplicative factor:

$$G_{\text{B}}(\mathbf{x}', \mathbf{x}; \Delta\tau) = \exp \left[-\left(\frac{V(\mathbf{x}) + V(\mathbf{x}')}{2} - E_T \right) \Delta\tau \right]. \quad (3.9)$$

In a DMC simulation, long imaginary-time propagation ($\tau = n\Delta\tau \rightarrow \infty$) is achieved by repeatedly applying this two-step process to an ensemble of configuration walkers: first shifting their coordinates via Gaussian diffusion, and subsequently replicating or destroying walkers according to the weight G_{B} .

3.2 Importance Sampling

To reduce statistical variance and improve computational efficiency, we introduce the importance sampling framework using an analytical trial wavefunction $\Psi_T(\mathbf{x})$.

3.2.1 The Quantum Drift Force

The standard DMC scheme presented in the previous section exhibits large statistical variance due to potential energy fluctuations. To mitigate this, importance sampling is introduced by guiding the random walk with an analytical trial wave function $\Psi_T(\mathbf{x})$. Instead of sampling $\phi(\mathbf{x}, \tau)$, the algorithm samples the mixed probability distribution $f(\mathbf{x}, \tau) \equiv \phi(\mathbf{x}, \tau)\Psi_T(\mathbf{x})$. Multiplying Eq. (3.6) by $\Psi_T(\mathbf{x})$ yields the modified evolution equation:

$$-\frac{\partial f(\mathbf{x}, \tau)}{\partial \tau} = -D \sum_{i=1}^N \frac{\partial^2 f(\mathbf{x}, \tau)}{\partial x_i^2} + D \sum_{i=1}^N \frac{\partial}{\partial x_i} [f(\mathbf{x}, \tau) F_i(\mathbf{x})] + [E_L(\mathbf{x}) - E_T] f(\mathbf{x}, \tau), \quad (3.10)$$

where $E_L(\mathbf{x}) \equiv \frac{H\Psi_T(\mathbf{x})}{\Psi_T(\mathbf{x})}$ is the local energy and $F_i(\mathbf{x}) \equiv 2\frac{\partial \ln |\Psi_T(\mathbf{x})|}{\partial x_i}$ is the 1D quantum drift force acting on particle i . The drift force acts as an effective velocity field driving walkers toward regions where Ψ_T^2 is large, drastically reducing sampling in nodal regions or high-potential barriers. The local energy $E_L(\mathbf{x})$ replaces $V(\mathbf{x})$ in the branching term. In the short-time approximation, the importance-sampled Green's function factorizes as $\tilde{G}(\mathbf{x}', \mathbf{x}; \Delta\tau) \approx \tilde{G}_{\text{diff}}(\mathbf{x}', \mathbf{x}; \Delta\tau) \tilde{G}_{\text{B}}(\mathbf{x}', \mathbf{x}; \Delta\tau)$. Assuming the drift force $\mathbf{F}(\mathbf{x})$ remains constant over a small time step $\Delta\tau$, the drift-diffusion kernel corresponds to a shifted Gaussian:

$$\tilde{G}_{\text{diff}}(\mathbf{x}', \mathbf{x}; \Delta\tau) = (4\pi D\Delta\tau)^{-N/2} \exp \left[-\frac{[\mathbf{x}' - \mathbf{x} - D\Delta\tau \mathbf{F}(\mathbf{x})]^2}{4D\Delta\tau} \right], \quad (3.11)$$

while the modified branching weight is governed by the local energy:

$$\tilde{G}_B(\mathbf{x}', \mathbf{x}; \Delta\tau) = \exp \left[- \left(\frac{E_L(\mathbf{x}) + E_L(\mathbf{x}')}{2} - E_T \right) \Delta\tau \right]. \quad (3.12)$$

Together, these terms bias the walker population toward energetically favorable regions while reducing the variance of the stochastic sampling.

3.2.2 Choice of Trial Wavefunction

To maximize algorithm efficiency and minimize statistical variance, the trial wavefunction Ψ_T should closely approximate the ground state Φ_0 . For a binary mixture of 1D Bose gases with inter- and intra-species contact interactions, an accurate choice consists in considering the pair-product Jastrow form of the trial wavefunction [27, 33]:

$$\Psi_T(\mathbf{x}) = \prod_{1 \leq i < j}^{N_\uparrow} f_{\uparrow\uparrow}(|x_i - x_j|) \prod_{1 \leq i < j}^{N_\downarrow} f_{\downarrow\downarrow}(|y_i - y_j|) \prod_{i=1}^{N_\uparrow} \prod_{j=1}^{N_\downarrow} f_{\uparrow\downarrow}(|x_i - y_j|), \quad (3.13)$$

where x_i and y_j denote the coordinates of component $\uparrow$ and $\downarrow$ particles, respectively. For simplicity, we chose the two-body correlation functions as

$$f_{\alpha\beta}(x) = \cos(k_{\alpha\beta}x + \delta_{\alpha\beta}); \quad \alpha, \beta \in \{\uparrow, \downarrow\}, \quad (3.14)$$

so that each Jastrow term corresponds to a solution of the two-body scattering problem. The momenta $k_{\alpha\beta}$ and phase shifts $\delta_{\alpha\beta}$ are determined by applying appropriate boundary conditions. First, periodic boundary conditions $\Psi_T(\dots, x_i + L, \dots) = \Psi_T(\dots, x_i, \dots)$ impose $f'_{\alpha\beta}(L/2) = 0$. Second, the Bethe-Peierls [32] boundary condition models zero-range contact interactions via a derivative cusp as $x \rightarrow 0$:

$$\left. \frac{f'_{\alpha\beta}(x)}{f_{\alpha\beta}(x)} \right|_{x=0} = -\frac{1}{a_{\alpha\beta}}, \quad (3.15)$$

where $a_{\alpha\beta} = -2\hbar^2/(mg_{\alpha\beta})$ is the 1D effective scattering length for coupling constant $g_{\alpha\beta}$. Combining these conditions yields the system of equations:

$$k_{\alpha\beta} \frac{L}{2} + \delta_{\alpha\beta} = 0, \quad (3.16)$$

$$\tan(\delta_{\alpha\beta}) = \frac{1}{k_{\alpha\beta} a_{\alpha\beta}}. \quad (3.17)$$

This system reduces to a transcendental equation for $\delta_{\alpha\beta}$ that is readily solved using standard root-finding algorithms (e.g., Brent's method [34]), after which $k_{\alpha\beta}$ is recovered directly. This trial wave function is well suited for simulating the uncoupled mixture, performing particularly well in both the weakly and strongly interacting regimes. Although it does not explicitly account for Rabi coupling, it will serve as a suitable option for the coupled mixture, and the branching process will correct any deficiencies in the trial state.

3.2.3 Forward Walking

While importance sampling reduces the variance in the observables, the sampled probability distribution, $f(\mathbf{x}) = \Psi_T(\mathbf{x})\Phi_0(\mathbf{x})$, still depends on a specific choice of the trial function and contains a certain bias. Consequently, the sampled averages are not taken directly over the exact ground-state distribution, but instead involve the trial wave function Ψ_T , yielding the *mixed estimator*,

$$\langle \hat{O} \rangle_{\text{mixed}} = \frac{\langle \Psi_T | \hat{O} | \Phi_0 \rangle}{\langle \Psi_T | \Phi_0 \rangle}. \quad (3.18)$$

Still, the mixed estimator for the energy is exact ($E_{\text{mixed}} = E_0$). However, observables which do not commute with the Hamiltonian exhibit systematic bias [35] toward the trial wave function Ψ_T . Obtaining exact ground-state expectation values requires the *pure estimator*,

$$\langle \hat{O} \rangle_{\text{pure}} = \frac{\langle \Phi_0 | \hat{O} | \Phi_0 \rangle}{\langle \Phi_0 | \Phi_0 \rangle} = \frac{\left\langle \Phi_0 \left| \hat{O} \frac{\Phi_0}{\Psi_T} \right| \Psi_T \right\rangle}{\left\langle \Phi_0 \left| \frac{\Phi_0}{\Psi_T} \right| \Psi_T \right\rangle}. \quad (3.19)$$

Equation (3.19) shows that mixed-distribution averages can be converted to pure expectations by weighting each walker configuration by $\Phi_0(\mathbf{x})/\Psi_T(\mathbf{x})$. Following Liu, Kalos, and Chester [36], this ratio is determined stochastically via forward walking: over a sufficiently long imaginary-time interval $\Delta\tau_{\text{fw}}$, branching projects out excited-state components, leaving the asymptotic number of descendants $W(\mathbf{x}_w)$ proportional to the wave function ratio, i.e., $W(\mathbf{x}_w) \propto \frac{\Phi_0(\mathbf{x}_w)}{\Psi_T(\mathbf{x}_w)}$. The pure estimator for an operator $\hat{O}$ is then evaluated as:

$$\langle \hat{O} \rangle_{\text{pure}} = \frac{\sum_w \hat{O}(\mathbf{x}_w) W(\mathbf{x}_w)}{\sum_w W(\mathbf{x}_w)}. \quad (3.20)$$

We track descendant weights using ancestor tagging [37]. At the start of a measurement window, active physical configurations are archived in a pending registry and assigned unique ancestor IDs. During branching, replicated descendants inherit the exact ancestor ID of their parent walker. After the propagation delay $\Delta\tau_{\text{fw}}$, Eq. (3.20) is evaluated on the archived configurations, weighted by their surviving descendant counts W_w .

3.3 Stochastic Spin-Flip Sampling

Having established the importance-sampled DMC framework, we now develop the algorithmic extensions required to simulate single-particle spin-flips.

3.3.1 Single-Particle Green's Function with Spin Tunneling

To solve the single-particle imaginary-time evolution under internal state coupling, we consider the diffusion problem:

$$-\frac{\partial \Psi(x, \sigma; \tau)}{\partial \tau} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi(x, \sigma; \tau)}{\partial x^2} - t \Psi(x, -\sigma; \tau) - E_T \Psi(x, \sigma; \tau), \quad (3.21)$$

where $\sigma = \pm 1$ represents the internal spin component ($\uparrow \rightarrow +1$, $\downarrow \rightarrow -1$). For an initial localized state $\Psi(x, \sigma; 0) = \delta(x - x_0)\delta_{\sigma,1}$, we decouple the tunneling term by mapping onto symmetric (+) and antisymmetric (-) eigenstates, $\Psi_{\pm}(x, \sigma; 0) = \delta(x - x_0)\frac{\delta_{\sigma,1 \pm \delta_{\sigma,-1}}}{2}$. Decomposing the initial state as $\Psi(x, \sigma; 0) = \Psi_+(x, \sigma; 0) + \Psi_-(x, \sigma; 0)$ splits Eq. (3.21) into two uncoupled equations:

$$-\frac{\partial \Psi_{\pm}}{\partial \tau} = \left[-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} - (E_T \pm t) \right] \Psi_{\pm}. \quad (3.22)$$

Each eigenstate evolves independently as a product of free spatial diffusion and exponential tunneling evolution:

$$\Psi_{\pm}(x, \sigma; \Delta\tau) = G_0(x, x_0; \Delta\tau) e^{(E_T \pm t)\Delta\tau} \left(\frac{\delta_{\sigma,1} \pm \delta_{\sigma,-1}}{2} \right), \quad (3.23)$$

where $G_0(x, x_0; \Delta\tau) = \left(\frac{m}{2\pi\hbar^2\Delta\tau} \right)^{1/2} \exp \left[-\frac{m(x-x_0)^2}{2\hbar^2\Delta\tau} \right]$ is the standard 1D free-particle Green's function [38]. Recombining $\Psi = \Psi_+ + \Psi_-$ yields the exact single-particle propagator:

$$G(x, \sigma; x_0, 1; \Delta\tau) = G_0(x, x_0; \Delta\tau) e^{E_T\Delta\tau} [\cosh(t\Delta\tau)\delta_{\sigma,1} + \sinh(t\Delta\tau)\delta_{\sigma,-1}]. \quad (3.24)$$

3.3.2 Drift and Diffusion in the Spin Direction

To incorporate tunneling into importance-sampled DMC method, we apply $f(x, \sigma, \tau) = \Psi_T(x, \sigma)\Psi(x, \sigma, \tau)$ to Eq. (3.21). Because the tunneling operator couples the discrete spin states $\sigma \in \{+1, -1\}$, evaluating $\Psi_T \frac{\partial \Psi}{\partial \tau}$ yields a rate-equation term representing transitions between spin states:

$$\begin{aligned} \frac{\partial f(x, \sigma, \tau)}{\partial \tau} &= D \frac{\partial^2 f(x, \sigma, \tau)}{\partial x^2} - D \frac{\partial}{\partial x} [f(x, \sigma, \tau) F(x, \sigma)] \\ &\quad + t \frac{\Psi_T(x, \sigma)}{\Psi_T(x, -\sigma)} f(x, -\sigma, \tau) - t \frac{\Psi_T(x, -\sigma)}{\Psi_T(x, \sigma)} f(x, \sigma, \tau) \\ &\quad + [E_T - E_L(x, \sigma)] f(x, \sigma, \tau), \end{aligned} \quad (3.25)$$

where the local energy includes the tunneling contribution:

$$E_L(x, \sigma) = -D \frac{1}{\Psi_T(x, \sigma)} \frac{\partial^2 \Psi_T(x, \sigma)}{\partial x^2} - t \frac{\Psi_T(x, -\sigma)}{\Psi_T(x, \sigma)}. \quad (3.26)$$

The spatial drift and diffusion terms conserve $f(x, \sigma, \tau)$ within each spin state. The tunneling terms in Eq. (3.25) do not contribute to branching, as they conserve the total number of walkers $N(\tau) = \sum_{\sigma=\pm 1} \int f(x, \sigma, \tau) dx$, as confirmed by summing over both spin states:

$$\begin{aligned} \sum_{\sigma=\pm 1} \left. \frac{\partial f(x, \sigma, \tau)}{\partial \tau} \right|_{\text{tunnel.}} &= \left[t \frac{\Psi_T(x, 1)}{\Psi_T(x, -1)} f(x, -1, \tau) - t \frac{\Psi_T(x, -1)}{\Psi_T(x, 1)} f(x, 1, \tau) \right] \\ &\quad + \left[t \frac{\Psi_T(x, -1)}{\Psi_T(x, 1)} f(x, 1, \tau) - t \frac{\Psi_T(x, 1)}{\Psi_T(x, -1)} f(x, -1, \tau) \right] = 0. \end{aligned} \quad (3.27)$$

To simulate this discrete Markov process over a time step $\Delta\tau$, consider a single walker initialized in spin state σ at $\tau = 0$, that is having $f(\sigma, 0) = 1$ and $f(-\sigma, 0) = 0$. Probability conservation $f(\sigma, \tau) + f(-\sigma, \tau) = 1$ yields a first-order rate equation for the occupation of state $-\sigma$,

$$\frac{\partial f(-\sigma, \tau)}{\partial \tau} = A - (A + B)f(-\sigma, \tau), \quad (3.28)$$

where $A = t \frac{\Psi_T(x, -\sigma)}{\Psi_T(x, \sigma)}$ and $B = t \frac{\Psi_T(x, \sigma)}{\Psi_T(x, -\sigma)}$. Integrating from $f(-\sigma, 0) = 0$ yields:

$$\begin{aligned} f(-\sigma, \Delta\tau) &= \frac{A}{A+B} \left(1 - e^{-(A+B)\Delta\tau} \right) \\ &= \frac{\Psi_T^2(x, -\sigma)}{\Psi_T^2(x, \sigma) + \Psi_T^2(x, -\sigma)} \left(1 - \exp \left\{ -t \left[\frac{\Psi_T(x, -\sigma)}{\Psi_T(x, \sigma)} + \frac{\Psi_T(x, \sigma)}{\Psi_T(x, -\sigma)} \right] \Delta\tau \right\} \right). \end{aligned} \quad (3.29)$$

For short imaginary-time steps ($t\Delta\tau \ll 1$), expanding $1 - e^{-y} \approx y$ reduces the exact solution to a simple stochastic transition rule,

$$P(\sigma \rightarrow -\sigma) = f(-\sigma, \Delta\tau) \approx \frac{\Psi_T(x, -\sigma)}{\Psi_T(x, \sigma)} t\Delta\tau. \quad (3.30)$$

This probability directly governs the rate of discrete spin-flip transitions during each Monte Carlo propagation step.

3.4 Observables

This section outlines the physical observables computed within our algorithm, which serve to characterize the ground-state properties and critical behavior of our system.

3.4.1 Energy

The ground-state energy E_0 can be accessed either through walker population dynamics or local energy averages. The growth estimator $E_g = E_T + \frac{1}{\tau_2 - \tau_1} \ln [N_w(\tau_1)/N_w(\tau_2)]$ evaluates walker population changes over imaginary time. While E_g is used dynamically to adjust E_T for population control, its large stochastic variance makes the local energy estimator a much better option for data collection.

Because importance sampling generates walker configurations distributed according to $f(\mathbf{x}) = \Psi_T(\mathbf{x})\Phi_0(\mathbf{x})$, the ensemble average of the local energy yields:

$$\langle E_L \rangle_f = \frac{\int \Phi_0(\mathbf{x}) H \Psi_T(\mathbf{x}) d\mathbf{x}}{\int \Psi_T(\mathbf{x}) \Phi_0(\mathbf{x}) d\mathbf{x}} = \frac{\langle \Phi_0 | H | \Psi_T \rangle}{\langle \Phi_0 | \Psi_T \rangle} = E_0, \quad (3.31)$$

where the Hermiticity of H guarantees an exact, unbiased estimate of E_0 . Furthermore, as $\Psi_T \rightarrow \Phi_0$, spatial fluctuations in $E_L(\mathbf{x})$ vanish, satisfying the zero-variance principle.

3.4.2 Moments of the Polarization

To evaluate magnetic order, we compute three polarization moments: $\langle |\hat{P}| \rangle$, $\langle \hat{P}^2 \rangle$, and $\langle \hat{P}^4 \rangle$. Because the polarization operator does not commute with the Hamiltonian,

these expectation values have to be evaluated as pure estimators via forward walking to eliminate trial wave function bias. Due to the system's $\mathbb{Z}_2$ spin-reversal symmetry, an ergodic Monte Carlo trajectory samples both symmetry-broken states, causing the direct order parameter average to vanish ($\langle \hat{P} \rangle = 0$). We therefore rely on the magnitude $\langle |\hat{P}| \rangle$ and second moment $\langle \hat{P}^2 \rangle$ to quantify magnetic ordering. These moments directly yield the polarization susceptibility [39],

$$\chi_P = N \left(\langle \hat{P}^2 \rangle - \langle |\hat{P}| \rangle^2 \right), \quad (3.32)$$

which measures order-parameter fluctuations. To locate the phase boundary without finite-size scaling ambiguities, we compute the Binder cumulant [40, 41, 42],

$$U_4 = 1 - \frac{\langle \hat{P}^4 \rangle}{3 \langle \hat{P}^2 \rangle^2}. \quad (3.33)$$

Binder cumulant U_4 is dimensionless and scale-invariant at criticality. That is, different curves for U_4 as a function of the control parameter, calculated for different system sizes N , intersect at a common crossing point, pinpointing the critical inter-species [43, 44].

3.4.3 Static Structure Factors

To probe spatial and spin ordering, we compute the static density structure factor $S_\rho(k)$ and spin structure factor $S_s(k)$ [17, 45],

$$S_\rho(k) = \frac{1}{N} \left\langle \left| \sum_{j=1}^N e^{ikx_j} \right|^2 \right\rangle, \quad S_s(k) = \frac{1}{N} \left\langle \left| \sum_{j=1}^N \sigma_j e^{ikx_j} \right|^2 \right\rangle, \quad (3.34)$$

where index $\sigma_j \in \{+1, -1\}$ denotes the spin component, and momentum $k = 2\pi n/L$ ($n \in \mathbb{Z}$) satisfies periodic boundary conditions. Mathematically, the static structure factors are related to the Fourier transforms of the pair density and spin-spin correlation functions [33], directly mapping two-body spatial order in momentum space. Physically, the density structure factor $S_\rho(k)$ measures total density modulations and manifests a linear phononic behavior at small momenta, whereas the spin structure factor $S_s(k)$ captures relative spin fluctuations and signals magnetic ordering or phase separation. Because these spatial correlation operators do not commute with our Hamiltonian 2.3 both observables have to be evaluated using pure estimators via forward walking to eliminate trial wave function bias.

3.5 DMC Stochastic Realization

Each walker $w \in \{1, \dots, N_w\}$ carries continuous spatial coordinates $\mathbf{x}_w = (x_{w,1}, \dots, x_{w,N})$, discrete spin variables $\mathbf{s}_w = (\sigma_{w,1}, \dots, \sigma_{w,N})$ with $\sigma_{w,i} \in \{+1, -1\}$, and an integer ancestor ID $\text{id}_w \in \mathbb{N}$. The initial ensemble is prepared with a starting polarization chosen as the clos-

est discrete value to the mean-field prediction 2.8, while the initial trial energy is set from a preparatory Variational Monte Carlo (VMC) [14, 15] calculation using this polarization. An equilibration phase of N_{eq} steps then relaxes the ensemble into $f(\mathbf{x}, \mathbf{s}) = \Psi_T(\mathbf{x}, \mathbf{s})\Phi_0(\mathbf{x}, \mathbf{s})$. Following equilibration, each time step $\Delta\tau$ proceeds as follows:

- **Drift-Diffusion Propagation:** Drift forces $\mathbf{F}$ and local energies E_L are evaluated. Continuous coordinates are updated on a periodic box of length L using either a first-order scheme $\mathcal{O}(\Delta\tau)$,

$$\mathbf{x}_{w,\text{new}} = \left(\mathbf{x}_w + 2D\mathbf{F}_{\text{old}}\Delta\tau + \boldsymbol{\eta}\sqrt{2D\Delta\tau} \right) \bmod L, \quad (3.35)$$

or a second-order predictor-corrector scheme $\mathcal{O}(\Delta\tau^2)$,

$$\mathbf{x}_{\text{pred}} = \left(\mathbf{x}_w + 2D\mathbf{F}_{\text{old}}\Delta\tau + \boldsymbol{\eta}\sqrt{2D\Delta\tau} \right) \bmod L, \quad (3.36)$$

$$\mathbf{x}_{w,\text{new}} = \left(\mathbf{x}_w + D(\mathbf{F}_{\text{old}} + \mathbf{F}_{\text{pred}})\Delta\tau + \boldsymbol{\eta}\sqrt{2D\Delta\tau} \right) \bmod L. \quad (3.37)$$

Local energies E_{new} and drift forces $\mathbf{F}_{\text{new}}$ are updated at $\mathbf{x}_{w,\text{new}}$.

- **Spin Tunneling:** For each particle $i = 1, \dots, N$, candidate spin flips $\sigma_{w,i} \rightarrow -\sigma_{w,i}$ are evaluated sequentially using the amplitude ratio $R_i = \Psi_T(\dots, -\sigma_i, \dots) / \Psi_T(\dots, \sigma_i, \dots)$ and transition probability (3.30). Accepted flips update local drift forces and energies immediately before evaluating particle $i + 1$.
- **Branching:** The branching weight w_k for walker k is given by:

$$w_k = \begin{cases} \exp[-\Delta\tau(E_{\text{old}} - E_T)], & \text{First-order} \\ \exp\left[-\Delta\tau\left(\frac{E_{\text{old}} + E_{\text{new}}}{2} - E_T\right)\right], & \text{Second-order} \end{cases} \quad (3.38)$$

Walker k is replicated into $M_k = \min(\lfloor w_k + \gamma \rfloor, M_{\text{max}})$ copies ($\gamma \sim U(0, 1)$). Replicated or deleted walkers duplicate or remove their associated ancestor ID id_k .

- **Population Control:** The reference energy E_T is updated to stabilize the population around $N_{w,0}$ (initial/target number of walkers):

$$E_T = \bar{E}_{\text{loc}} - \frac{1}{\Delta\tau} \ln \left(\frac{N_w}{N_{w,0}} \right). \quad (3.39)$$

- **Ancestor ID and Pure Observable Accumulation:** When a forward-walking window matures ($\Delta\tau_{\text{fw}} = M\Delta\tau$), descendant weights W_w are computed by counting active walkers carrying each archived ancestor ID and pure observables are evaluated via Eq. (3.20). At designated intervals, new windows are initialized by archiving the current configurations and assigning unique ancestor IDs $\text{id}_w = w$.
- **Mixed Observable Accumulation:** The ensemble-averaged local energy $\bar{E}_{\text{loc}} = \frac{1}{N_w} \sum_{w=1}^{N_w} E_L(\mathbf{x}_w, \mathbf{s}_w)$ is accumulated as the mixed energy estimator.

Chapter 4

Results

This chapter presents DMC results, focusing on finite-size effects and static structure factors, with quantitative validation against the exact two-body solution and mean-field and beyond-mean-field limits.

4.1 Finite-Size Scaling

Phase transitions are thermodynamic phenomena defined by non-analyticities that only emerge in the infinite-system limit ($N \rightarrow \infty$) [46]. In finite systems, these non-analyticities are smoothed into rounded crossovers. However, true thermodynamic behavior can be systematically obtained by analyzing the system behavior seen when increasing particle numbers N . For example, as shown in panel a) of Fig. 4.1, in the unpolarized phase, the polarization $\langle |\hat{P}| \rangle$ remains finite, unlike the mean-field prediction, $\langle \hat{P} \rangle = 0$, because the absolute value prevents quantum fluctuations from canceling as polarization has a discrete spectrum spaced by $2/N$. These finite-size artifact fluctuations decay systematically as N increases, transforming the broad crossover into a sharp phase transition.

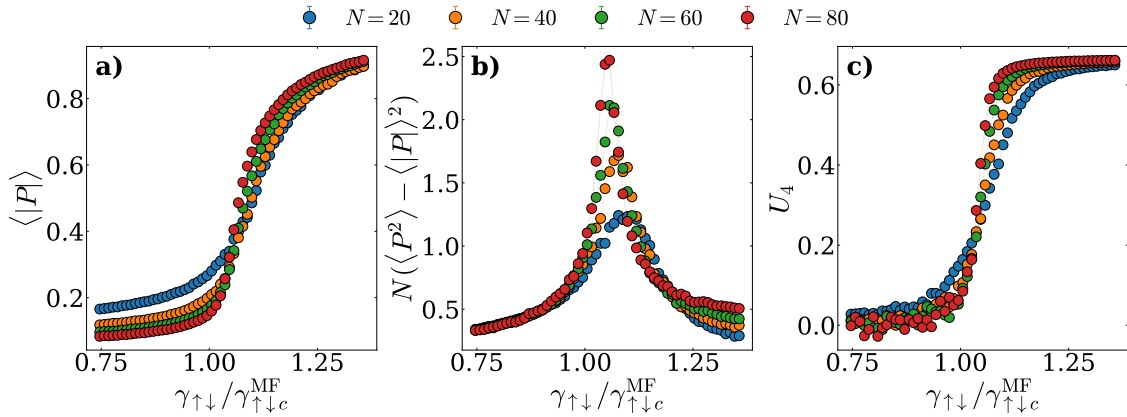


Figure 4.1: a) Absolute polarization, b) Fluctuations of the polarization, c) Binder cumulant. The horizontal axis shows inter-species interaction strength, normalized to the mean-field critical value in 2.2.2. Simulation parameters are set to $\gamma = 0.01$ and $\tilde{t} = 0.125\gamma$ and correspond to the weakly interacting regime.

Next, order-parameter fluctuations (panel (b) of Fig. 4.1) are expected to diverge at criticality in the thermodynamic limit. In finite systems, however, this formal divergence

rounds into a peak displaced from the thermodynamic critical interaction. As N increases, this peak grows taller, narrows and shifts systematically toward the thermodynamic critical value, allowing extrapolation via finite-size scaling. Alternatively, the thermodynamic critical point can be pinpointed using Binder cumulant crossings, as U_4 is scale-invariant at criticality, curves for different system sizes intersect at a single point (Figure 4.1, panel c).

4.2 Static Structure Factor

Figure 4.2 presents DMC results for the density $S_\rho(k)$ and spin $S_s(k)$ static structure factors across the quantum phase transition, displaying qualitatively distinct behavior.

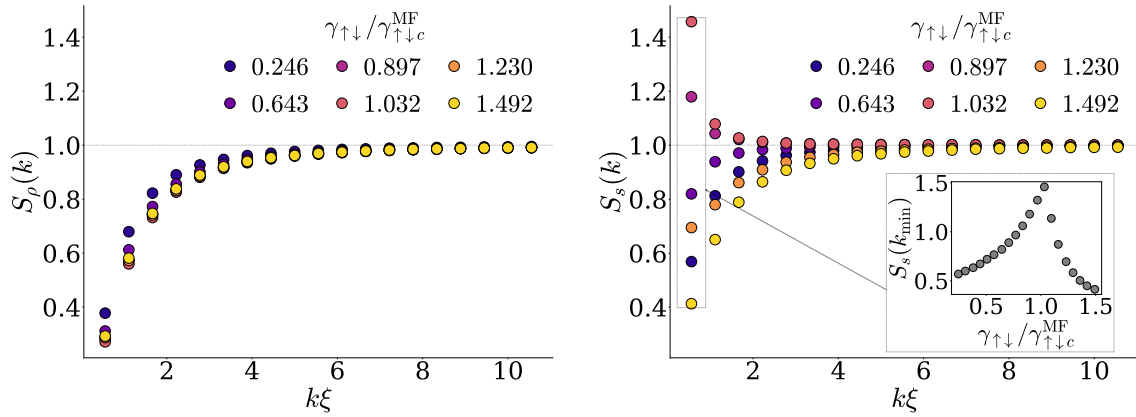


Figure 4.2: Static density $S_\rho(k)$ (left) and spin $S_s(k)$ (right) structure factors as a function of wavenumber k (normalized to ξ^{-1} , with $\xi = \hbar/\sqrt{2mgn}$) across the quantum phase transition. Inset: Low-momentum peak value $S_s(k_{\min})$ (with $k_{\min} = 2\pi/L$) vs. normalized interspecies interaction strength $\gamma_{\uparrow\downarrow}/\gamma_{\uparrow\downarrow c}^{\text{MF}}$ for $\gamma = 0.01$, $\tilde{t} = 0.125\gamma$, $N = 80$.

The density structure factor $S_\rho(k)$ (Figure 4.2, left) displays a linear low-momentum acoustic limit, $S_\rho(k) \propto k$ as $k \rightarrow 0$. This behavior directly reflects the gapless, acoustic nature of the density excitation branch $E_{+,k} \approx c_d|k|$, which governs the long-wavelength limit via the Feynman single-mode approximation: $S_\rho(k) = \frac{\hbar^2 k^2}{2mE_{+,k}} \approx \left(\frac{\hbar}{2mc_d}\right)|k|$. Across all interaction strengths $\gamma_{\uparrow\downarrow}$, $S_\rho(k)$ remains smooth and featureless, confirming that total-density fluctuations remain phonon-like and do not participate in driving the transition.

In contrast, the spin structure factor $S_s(k)$ (Figure 4.2, right) captures the mechanism of the polarizing phase transition. Well below the transition region ($\gamma_{\uparrow\downarrow} \ll \gamma_{\uparrow\downarrow c}^{\text{MF}}$), $S_s(k)$ remains finite as $k \rightarrow 0$ due to the gapped nature of the spin excitations. Upon increasing $\gamma_{\uparrow\downarrow}$, $S_s(k)$ develops a prominent low-momentum peak. As shown in the inset, tracking the lowest accessible momentum component $S_s(k_{\min})$ reveals a sharp maximum near the transition region. Because long-wavelength spin fluctuations govern the order-parameter variance, this low-momentum enhancement offers a momentum-space realization of the critical polarization fluctuations previously identified in Figure 4.1 (b). Well into the polarized regime, relative-density fluctuations are strongly suppressed and $S_s(k)$ collapses back toward the single-component density structure factor $S_\rho(k)$.

4.3 Two-Body Problem

A first quantitative benchmark of our DMC implementation is provided by comparing against the exact two-body solution. Since the trial wave function we use is based on an uncoupled system and does not explicitly contain tunneling, this setup is well suited for testing the DMC: branching must correct the trial state to yield exact ground-state energies, while forward walking must recover unbiased spin observables.

Because spin-flip symmetry makes the linear estimator $\langle \hat{P} \rangle = P_{\uparrow\uparrow} - P_{\downarrow\downarrow}$ vanish, we instead compute $\langle \hat{P}^2 \rangle = P_{\uparrow\uparrow} + P_{\downarrow\downarrow}$, which receives contributions only from same-spin pairs. In Fig. 4.3 (left panel), DMC shows excellent agreement with the exact energy density across all interaction regimes. The right panel confirms excellent agreement for $\langle \hat{P}^2 \rangle$, validating the forward walking method. Its inset shows convergence as forward walking time $\Delta\tau_{\text{FW}} \rightarrow \infty$, successfully eliminating substantial trial wave function bias.

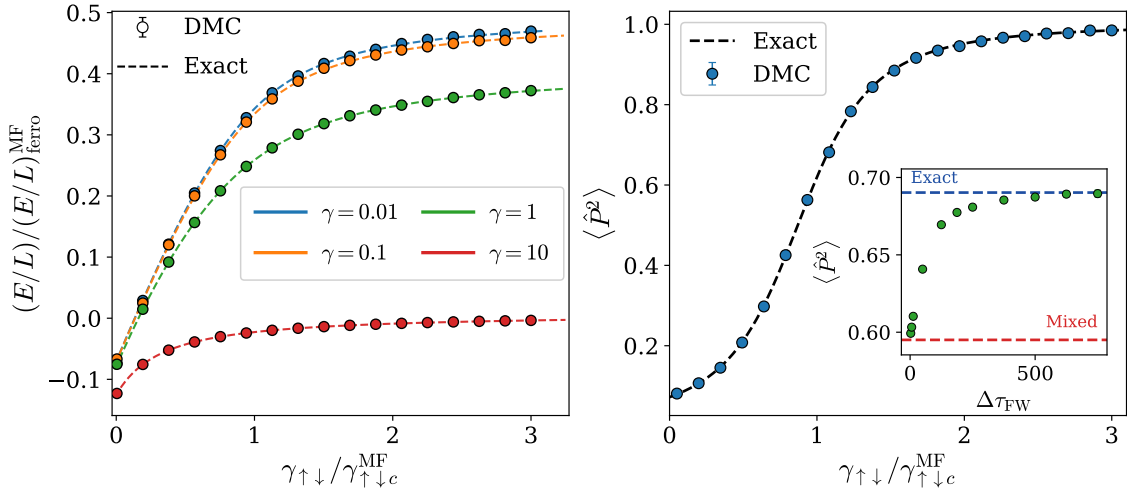


Figure 4.3: Left: Energy density (normalized to the mean-field fully polarized value $(E/L)_{\text{ferro}}^{\text{MF}}$) for various dimensionless intra-species interaction strengths γ . Right: Squared order parameter $\langle \hat{P}^2 \rangle$ at $\gamma = 0.01$ and $\tilde{t} = 0.1\gamma$. Inset: $\langle \hat{P}^2 \rangle$ vs forward walk window (in units of $mL^2/\hbar^2$). Both quantities plotted vs relative dimensionless inter-species interaction strength $\gamma_{\uparrow\downarrow}/\gamma_{\uparrow\downarrow c}^{\text{MF}}$. In both panels Markers are DMC results, dashed lines exact results.

4.4 Mean-Field and Beyond-Mean-Field Limits

The next stage of quantitative validation evaluates the ground-state energy density and the mean square order parameter in the weakly interacting regime against the mean-field (MF) and beyond-mean-field (BMF) predictions.

Energy density: In Fig. 4.4 we compare DMC results with MF and BMF predictions. In the unpolarized phase ($\gamma_{\uparrow\downarrow}/\gamma_{\uparrow\downarrow c}^{\text{MF}} < 1$ and inset a), the energy increases linearly with the inter-species interaction strength before smoothly transitioning toward a constant plateau, signaling the complete suppression of inter-species interactions characteristic of the fully polarized phase ($\gamma_{\uparrow\downarrow}/\gamma_{\uparrow\downarrow c}^{\text{MF}} \gg 1$ and inset c). The smooth crossover between these regimes (inset b) reflects the second-order nature of the transition.

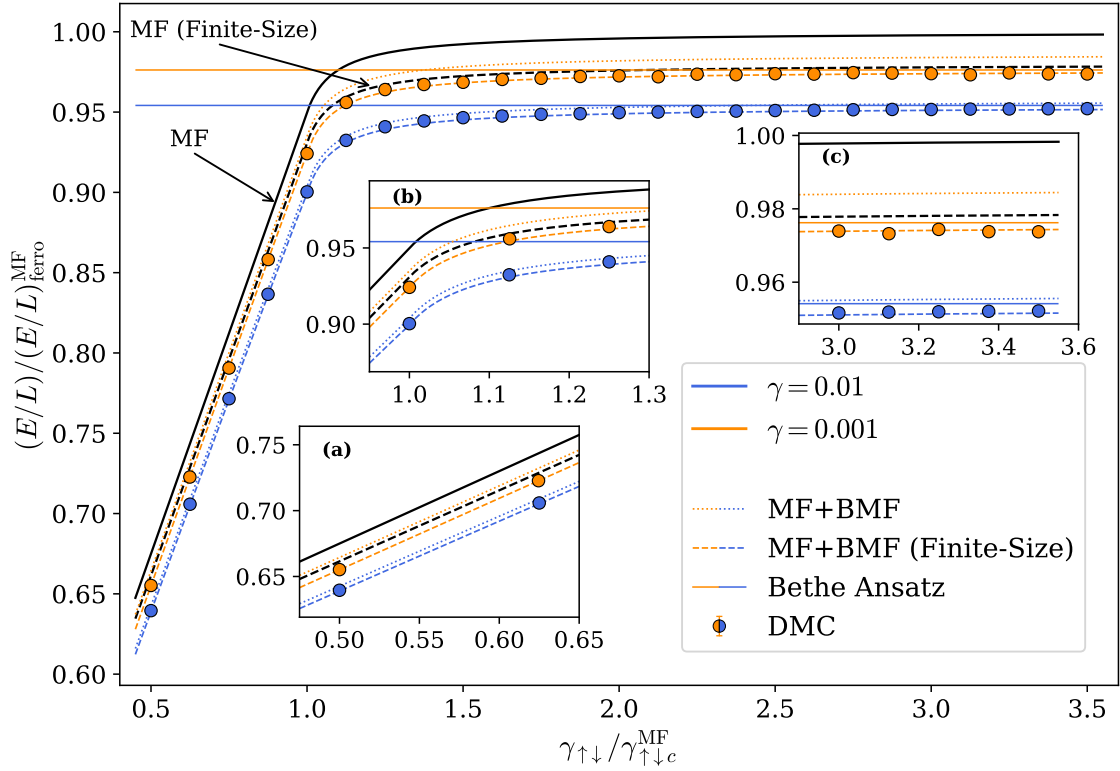


Figure 4.4: Energy density versus inter-species interaction strength (normalized to the MF critical value) for $N = 50$ and $\tilde{t} = 0.05\gamma$. Colors denote intra-species interaction strengths (γ). Curves show thermodynamic MF (solid black), finite-size MF (dashed black), thermodynamic BMF (dotted colored), finite-size BMF (dashed colored), and single-component Bethe Ansatz (solid colored) predictions compared to DMC data.

Across the entire inter-species spectrum, the DMC results show excellent quantitative agreement with the finite-size BMF energy density prediction for both values of γ . At large inter-species interaction strengths, both DMC and finite-size BMF results converge to the ground-state energy of a single-component Bose gas with equivalent intra-species interaction, a regime governed by the Lieb-Liniger model, as verified by the smooth approach to the exact Bethe-Ansatz [21, 25] limit. As intra-species interactions decrease (smaller γ), the energy progressively approaches the MF prediction, which lies above the BMF curve, confirming that beyond-mean-field effects lower the ground-state energy. For a finite number of particles, the energy approaches the thermodynamic MF result modified by a finite-size correction factor $(1 - 1/N)$. Finally, we observe that finite-size effects become increasingly pronounced at smaller γ ; this is particularly clear in inset c when comparing the DMC results directly against the thermodynamic BMF prediction which reflects the increasing healing length at weaker intra-species interactions.

Mean Square Order Parameter: We present a second quantitative comparison using the mean-square order parameter in Fig. 4.5. As discussed previously, two BMF corrections are

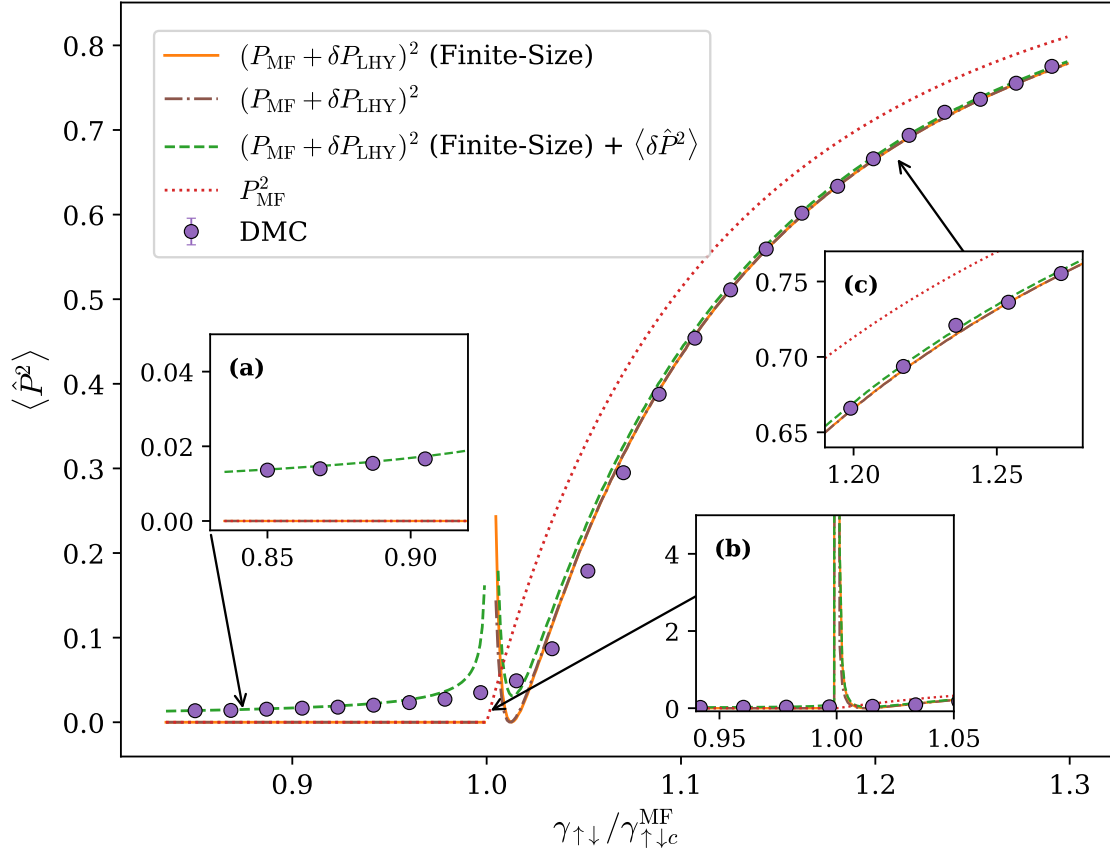


Figure 4.5: Mean-squared order parameter versus inter-species interaction strength (normalized to the MF critical value) for $N = 90$, $\gamma = 0.01$ and $\tilde{t} = 0.15\gamma$. Lines indicate mean-field (dotted red), beyond-mean-field due to LHY correction with finite-size (solid orange) and in the thermodynamic limit (point-dash brown) and beyond-mean-field with both LHY and quantum fluctuations (dashed green) predictions compared to DMC data (markers).

considered, arising from the LHY energy and quantum fluctuations. As shown in Fig 4.5, for 90 particles, finite-size and thermodynamic BMF predictions for $\langle \hat{P}^2 \rangle$ based on the LHY correction are nearly indistinguishable. In contrast, finite-size effects in the quantum-fluctuation correction remain visible through a non-zero polarization in the unpolarized phase. Comparison with the MF prediction further shows that beyond-mean-field effects reduce the polarization at a given inter-species interaction.

Deep in the unpolarized phase (inset a), DMC data show a finite polarization in excellent agreement with the quantum-fluctuation BMF correction, while MF and LHY-corrected MF predict zero polarization. At the mean-field critical point, both beyond-mean-field predictions diverge (inset b), an artifact of the Bogoliubov approximation. Consequently, the DMC results deviate from the BMF predictions near the critical point. This discrepancy is a finite-size effect that narrows with increasing particle number. Deeper in the polarized phase (inset c), the DMC results again show strong agreement with the BMF corrections.

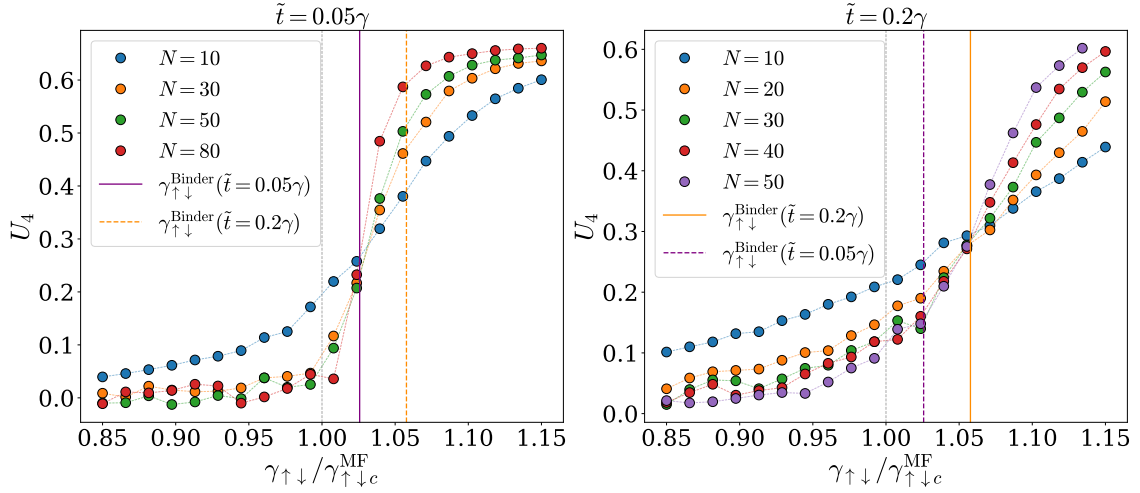


Figure 4.6: Binder cumulant versus normalized inter-species interaction strength for $\gamma = 0.01$ across system sizes at $t/gn = 0.05$ (left) and $t/gn = 0.2$ (right). Dotted gray line marks the mean-field critical point. Vertical purple ($t/gn = 0.05$) and yellow ($t/gn = 0.2$) lines show approximate Binder crossings, indicating critical point shifts of $\sim 3\%$ and $\sim 6\%$.

Critical point: We investigate the critical point using the method of Binder cumulant crossings. We find that there is a consistent shift of the critical point towards larger inter-species interactions compared with the MF prediction. In Fig. 4.6 we demonstrate that this positive shift increases with the tunneling strength $\tilde{t}/\gamma$. Indeed, an increase in t moves the mean-field critical point towards stronger inter-species interactions, at which BMF effects may become stronger. Similarly, at fixed $\tilde{t}/\gamma$, the shift increases with the intra-species interaction γ , resulting in an increasingly favored unpolarized phase in the strongly-interacting regime.

At the same time, the Binder cumulant crossings converge to the critical mean-field prediction in two particular regimes. First, for arbitrary Rabi coupling $\tilde{t}$, the mean-field result is recovered in the limit of very weak interactions, $\gamma \ll 1$. This is expected, as mean-field theory is generally valid in the weakly interacting regime, as further supported by Fig. 4.4, where decreasing γ brings the energy progressively closer to the mean-field prediction. On the other hand, the mean-field critical inter-species interaction strength remains valid for arbitrary intra-species interaction strengths γ in the ultraweak Rabi coupling limit $\tilde{t} \rightarrow 0$. According to mean-field theory, the critical phase boundary in this limit is $g_{\uparrow\downarrow c}^{\text{MF}} = g$. While this result may appear counter-intuitive and requires further investigation, it can be interpreted by recovering a well-known result for the uncoupled mixture ($t = 0$), introduced in Sec. 2.1.2, discussed in [27] and also available in Appendix B: the demixing transition occurs at all interaction-strength scales when the miscibility condition [24], $g_{\uparrow\downarrow} = g$, is satisfied. In the ultraweak coupling limit, the energy landscape may therefore remain effectively unaffected by the coupling and therefore inherited from the uncoupled case, while its breaking of the $\mathbb{Z}_2$ symmetry allows violations of individual component particle-number conservation. Further details can be found in Appendix C.

Chapter 5

Conclusions and Outlook

We have developed an exact Quantum Monte Carlo framework, based on the Diffusion Monte Carlo (DMC) technique with importance sampling and forward walking. To account for the internal state dynamics, the standard DMC scheme is extended with a stochastic step proposing spin-flip events.

The developed method has been implemented. We benchmarked the method against the exact two-body solution and theoretical predictions in both the mean-field and beyond-mean-field regimes, using the ground-state energy density and mean-square order parameter as primary observables. Excellent agreement is found with the two-body solution and beyond-mean-field predictions, while mean-field agreement is recovered asymptotically.

We find that the crossover between paramagnetic and ferromagnetic configurations, observed in a finite-size system, gets sharper as the number of particles is increased. Eventually, in the thermodynamic limit, a quantum phase transition is expected. We identify the spin sector as the mechanism driving the transition. In particular, this is evidenced by the long-wavelength divergence of the spin static structure factor while the density static structure factor vanishes in the limit of $k \rightarrow 0$.

Analysis of the critical point through Binder cumulant crossings reveals positive shifts in the critical inter-species interaction strength, which increase with interaction strength and systematically vanish in two limits: the weakly interacting regime ($\gamma \ll 1$) for arbitrary tunneling strength $\tilde{t}$, and the ultra-weak-coupling regime ($\tilde{t} \ll 1$) for arbitrary interaction strength γ . While the mean-field validity in the weakly interacting regime is expected, the ultra-weak-coupling limit is less intuitive. We speculate that this behavior arises from the inheritance of the energy landscape of the uncoupled mixture ($\tilde{t} = 0$), together with the activation of the polarizing transition by finite Rabi coupling $\tilde{t} > 0$.

This work lays the foundation for several promising research avenues. Immediate extensions include mapping the complete phase diagram across the full parameter space and quantification of the beyond-mean-field corrections. Another important question is to extract the critical exponents to verify their agreement or disagreement with the mean-field universality class. Future work could also explore the strongly interacting polarizing transition beyond the ultra-weak coupling limit, investigate asymmetric interaction configurations, incorporate external confining potentials, and extend the DMC framework to study these coherently coupled mixtures in higher dimensions.

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Appendix A

General Bogoliubov Excitations

To complement the excitation spectrum $E_{\pm}(k)$ presented in Section 2.2.3 for the unpolarized ground state ($n_{\uparrow} = n_{\downarrow} = n/2$), we provide here the dispersion relations valid for arbitrary component densities $n_{\uparrow}$ and $n_{\downarrow}$ (or polarization $P \neq 0$) following [28].

Minimizing the mean-field grand potential density $\Omega^{(0)}/L$ with respect to the classical fields $\psi_{\sigma} = \sqrt{n_{\sigma}}$ yields the species chemical potentials:

$$\mu_{\uparrow} = gn_{\uparrow} + g_{\uparrow\downarrow}n_{\downarrow} - t\sqrt{\frac{n_{\downarrow}}{n_{\uparrow}}}, \quad \mu_{\downarrow} = gn_{\downarrow} + g_{\uparrow\downarrow}n_{\uparrow} - t\sqrt{\frac{n_{\uparrow}}{n_{\downarrow}}}. \quad (\text{A.1})$$

Expanding the field operators around these condensate densities up to quadratic order in quantum fluctuations yields the Bogoliubov Hamiltonian matrix blocks:

$$A = \begin{pmatrix} \epsilon_k + gn_{\uparrow} + t\sqrt{\frac{n_{\downarrow}}{n_{\uparrow}}} & g_{\uparrow\downarrow}\sqrt{n_{\uparrow}n_{\downarrow}} - t \\ g_{\uparrow\downarrow}\sqrt{n_{\uparrow}n_{\downarrow}} - t & \epsilon_k + gn_{\downarrow} + t\sqrt{\frac{n_{\uparrow}}{n_{\downarrow}}} \end{pmatrix}, \quad B = \begin{pmatrix} gn_{\uparrow} & g_{\uparrow\downarrow}\sqrt{n_{\uparrow}n_{\downarrow}} \\ g_{\uparrow\downarrow}\sqrt{n_{\uparrow}n_{\downarrow}} & gn_{\downarrow} \end{pmatrix}, \quad (\text{A.2})$$

where $\epsilon_k = \frac{\hbar^2 k^2}{2m}$. Diagonalizing the corresponding 4×4 non-Hermitian eigenvalue problem $\begin{pmatrix} A & B \\ -B & -A \end{pmatrix} \mathbf{w} = E_{\lambda,k} \mathbf{w}$ yields the exact, general dispersion relations $E_{\pm,k}$ for arbitrary polarizations:

$$E_{\pm,k} = \sqrt{D_k \pm \sqrt{D_k^2 - \epsilon_k \left(\epsilon_k + t \frac{n_{\uparrow} + n_{\downarrow}}{\sqrt{n_{\uparrow}n_{\downarrow}}} \right) \mathcal{Q}_k}}, \quad (\text{A.3})$$

where D_k and $\mathcal{Q}_k$ are given by:

$$D_k = \frac{1}{2} \sum_{\sigma \in \{\uparrow, \downarrow\}} \left(\epsilon_k + t\sqrt{\frac{n_{\bar{\sigma}}}{n_{\sigma}}} \right) \left(\epsilon_k + 2gn_{\sigma} + t\sqrt{\frac{n_{\bar{\sigma}}}{n_{\sigma}}} \right) - t(2g_{\uparrow\downarrow}\sqrt{n_{\uparrow}n_{\downarrow}} - t), \quad (\text{A.4})$$

$$\mathcal{Q}_k = \prod_{\sigma \in \{\uparrow, \downarrow\}} \left(\epsilon_k + 2gn_{\sigma} + t\sqrt{\frac{n_{\bar{\sigma}}}{n_{\sigma}}} \right) - (2g_{\uparrow\downarrow}\sqrt{n_{\uparrow}n_{\downarrow}} - t)^2, \quad (\text{A.5})$$

with $\bar{\sigma}$ denoting the state opposite to σ .

Appendix B

The Demixing Phase Transition

For a symmetric mixture ($g_{\uparrow\uparrow} = g_{\downarrow\downarrow} = g$) with repulsive interactions ($g, g_{\uparrow\downarrow} > 0$), the mean-field energy density is given by Eq. (2.6) with $t = 0$. The ground state can either form a uniform mixture of the two components or segregate into a phase-separated configuration. In the uniform configuration, the energy density takes the form

$$\frac{E_{\text{unif}}}{L} = \frac{1}{2}g \left[\left(\frac{N_{\uparrow}}{L} \right)^2 + \left(\frac{N_{\downarrow}}{L} \right)^2 \right] + g_{\uparrow\downarrow} \left(\frac{N_{\uparrow}}{L} \right) \left(\frac{N_{\downarrow}}{L} \right), \quad (\text{B.1})$$

while in the phase-separated configuration, one finds

$$\frac{E_{\text{separ}}}{L} = \frac{1}{2}g \left[\left(\frac{N_{\uparrow}}{L_{\uparrow}} \right)^2 + \left(\frac{N_{\downarrow}}{L_{\downarrow}} \right)^2 \right], \quad (\text{B.2})$$

where $L_{\uparrow}$ and $L_{\downarrow}$ are the spatial lengths occupied by the two separated domains ($L_{\uparrow} + L_{\downarrow} = L$). Mechanical equilibrium between the two phases is ensured by pressure balance across the interface, $\partial E_{\text{separ}} / \partial L_{\uparrow} = \partial E_{\text{separ}} / \partial L_{\downarrow}$, which yields the condition $g(N_{\uparrow}/L_{\uparrow})^2 = g(N_{\downarrow}/L_{\downarrow})^2$ (or $g_{\uparrow\uparrow}(N_{\uparrow}/L_{\uparrow})^2 = g_{\downarrow\downarrow}(N_{\downarrow}/L_{\downarrow})^2$ for asymmetric mixtures). Substituting this back gives

$$\frac{E_{\text{separ}}}{L} = \frac{1}{2}g \left[\left(\frac{N_{\uparrow}}{L} \right)^2 + \left(\frac{N_{\downarrow}}{L} \right)^2 \right] + g \frac{N_{\uparrow}N_{\downarrow}}{L^2}. \quad (\text{B.3})$$

Comparing Eq. (B.3) and Eq. (B.1) shows that the stability condition required to prevent phase separation ($E_{\text{separ}} > E_{\text{unif}}$) is $g_{\uparrow\downarrow} < g$. For repulsive interactions, this condition also guarantees dynamic stability of the uniform phase against local density fluctuations (i.e. $[\partial^2 E / \partial N_{\uparrow}^2][\partial^2 E / \partial N_{\downarrow}^2] > [\partial^2 E / (\partial N_{\uparrow} \partial N_{\downarrow})]^2$). Analogously to what we do in section 2.2.2, we can rewrite equation B.2 as a function of the population imbalance or polarization $P = (N_{\uparrow} - N_{\downarrow})/N$, leading to

$$\frac{E(P)}{L} = \frac{n^2}{4} [(g + g_{\uparrow\downarrow}) + (g - g_{\uparrow\downarrow})P^2]. \quad (\text{B.4})$$

Equation B.4 allows us to determine the polarization sectors that minimize the ground-state

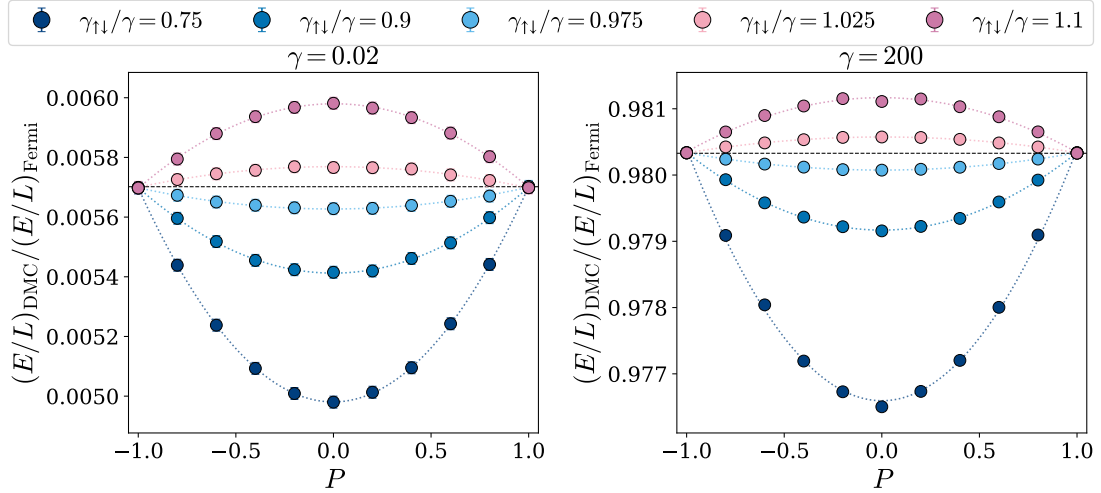


Figure B.1: Energy density of the mixture as a function of polarization for different values of dimensionless inter-species interaction strength normalized to dimensionless intra-species interaction strength. Left panel refers to $\gamma = 0.02$ and right panel refers to $\gamma = 200$. Energy densities are shown in units of $(E/L)_{\text{Fermi}} = \pi^2 n^3 \hbar / 6m$ corresponding to the Fermi energy density when $\gamma = \infty$. The dashed lines are the best fits quadratic in P to guide the eye. Dashed black line indicates energy of the fully polarized sectors.

energy density at a given inter-species interaction strength. For $g_{\uparrow\downarrow} < g$, the favored sector is $P = 0$, while for $g_{\uparrow\downarrow} > g$, the favored sectors are $|P| = 1$. Exactly at the demixing point, $g_{\uparrow\downarrow} = g$, all polarization sectors are degenerate in energy. Thus, the favored polarization sector changes abruptly across the demixing critical point, indicating the discontinuous change in the order parameter characteristic of a first-order phase transition.

The discussed abrupt behavior persists across all interaction strength scales, as shown in B.1 in which we can see that below the demixing point, the minimum of the energy is always the $P = 0$ sector while above the critical point, the minimum of the energy is the fully polarized sector $|P| = 1$. As shown, this happens for both weak and strongly-interacting systems. This was discussed in [27], where a susceptibility $\chi = [\partial(E/L)^2 / n \partial P^2]^{-1}$ is defined as a response function to describe the behavior previously explained. As shown in B.1, the energy density changes from a minimum at $P=0$ to two minima at $P = \pm 1$, exactly at criticality, the energy density as a function of polarization becomes flat with vanishing curvature, i.e. $\partial^2(E/L) / n \partial P^2 \rightarrow 0$, causing the susceptibility to diverge at the critical point.

Appendix C

The Ultraweak Coupling Limit

In the ultraweak coupling limit ($\tilde{t}/\gamma \ll 1$), the critical inter-species interaction predicted by mean-field theory remains remarkably robust across all interaction strengths. To understand this behavior, it is instructive to examine the underlying energy landscape in the absence of Rabi coupling ($t = 0$). At $t = 0$, a first-order demixing quantum phase transition occurs universally across all interaction scales, where the ground-state energy favors an abrupt shift from the unpolarized sector ($P = 0$) to a fully polarized state ($|P| = 1$) (see Appendix B). Because individual component particle numbers are strictly conserved at $t = 0$, polarization remains a static quantity during imaginary-time evolution. Introducing a weak Rabi coupling breaks this conservation law without significantly distorting the underlying energy landscape. Consequently, the system inherits the energy competition between unpolarized and fully polarized phases, leading to a sharp, first-order-like (though formally second-order) polarizing quantum phase transition across the entire interaction range.

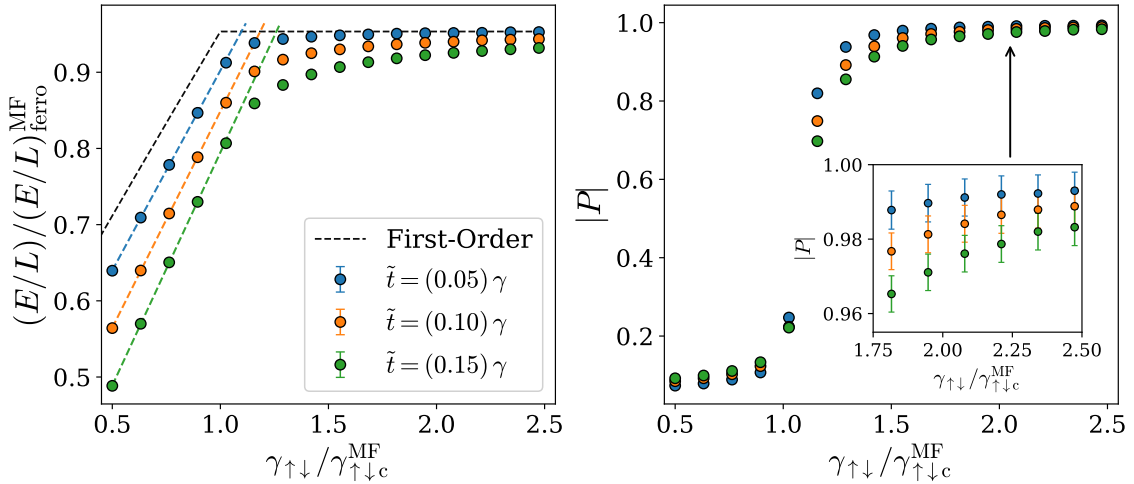


Figure C.1: (a) Energy density (normalized to MF fully polarized energy density, with the dashed line indicating the $t = 0$ level-crossing limit. (b) Absolute polarization. Both against dimensionless inter-species relative to MF prediction. Parameters: $\gamma = 0.01$, $N = 50$.

This mechanism directly accounts for the asymptotic first-order behavior observed as $\tilde{t}/\gamma \ll 1$, where the critical inter-species strength of the polarizing transition coincides with

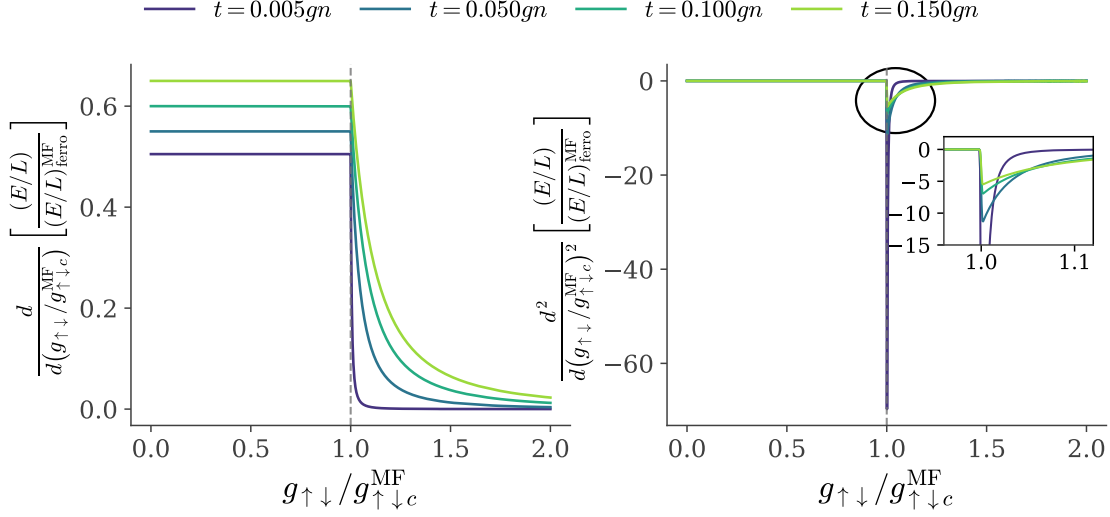


Figure C.2: Mean-field ground-state energy density first (left) and second (right) derivative with respect to inter-species interaction strength as a function of inter-species interaction strength. Energy density normalized to fully polarized phase mean-field energy density $(E/L)_{\text{ferro}}^{\text{MF}} = gn/2$ and inter-species interaction strength normalized to critical mean-field inter-species interaction strength. Multiple t/gn ratios plotted.

the miscibility criterion. As shown in Fig. C.1, reducing $\tilde{t}/\gamma$ causes the ground-state energy (panel a) to systematically approach a sharp level crossing or *kink* between the unpolarized ($P = 0$) and fully polarized ($|P| = 1$) phases. Concurrently, the order parameter (panel b) progressively sharpens, meaning the polarization at a given inter-species interaction strength increases as $\tilde{t}/\gamma$ is reduced. In the $\tilde{t}/\gamma \rightarrow 0$ limit, the order parameter looks like a near-discontinuous jump.

This behavior is also predicted by mean-field theory. The most straightforward way of seeing this is by taking $t \rightarrow 0$ in equation 2.8. By doing so, the result is, as expected, a discontinuous jump:

$$P = \begin{cases} 0, & \text{for } g_{\uparrow\downarrow} < g + \frac{2t}{n}, \\ \pm 1, & \text{for } g_{\uparrow\downarrow} > g + \frac{2t}{n}. \end{cases} \quad (\text{C.1})$$

In addition, by taking 2.7 and plugging the ground-state polarization 2.8 one can find the ground state energy density. Studying it together with the first and second derivative as $t \rightarrow 0$, the first-order asymptotic behavior is again revealed. This is shown in Figure C.2. In the left panel we see that as t/gn decreases the first derivative progressively approaches a discontinuous behavior at the critical point. This is the main signature of a first-order phase transition. Due to this asymptotic discontinuous behavior at criticality, in the right panel one can see that the second derivative diverges at the critical point as $t/gn \rightarrow 0$.